

Information and Entertainment System

 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)

 ST3093-A	Fluke 77-IV Digital Multimeter FLU77-4 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware or equivalent diagnostic scan tool
 ST2574-A	Flex Probe Kit NUD105-R025D or equivalent
 ST3104-A	Back-probe Pins POM6411 or equivalent
 ST3051-A	Universal Serial Bus (USB) Male-A To Male-A Cable CCMUSB2-AM-AM-10
 ST3085-A	Multi-Media Interface Tester 105-00120

DTC Charts

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

NOTE: For SYNC® related concerns, REFER to the SYNC® Symptom Chart before connecting the scan tool and beginning DTC-based diagnostics.

Audio Front Control Module (ACM) DTC Chart

DTC	Description	Action
B1A01:01	Speaker #1: General Electrical Failure	- For a L.F. door speaker concern, GO to Pinpoint Test F - For a L.F. tweeter speaker concern, GO to Pinpoint Test G
B1A01:11	Speaker #1: Circuit Short to Ground	- For a L.F. door speaker concern, GO to Pinpoint Test F - For a L.F. tweeter speaker concern, GO to Pinpoint Test G
B1A01:12	Speaker #1: Circuit Short to Battery	- For a L.F. door speaker concern, GO to Pinpoint Test F - For a L.F. tweeter speaker concern, GO to Pinpoint Test G
B1A01:13	Speaker #1: Circuit Open	- For a L.F. door speaker concern, GO to Pinpoint Test F - For a L.F. tweeter speaker concern, GO to Pinpoint Test G
B1A02:01	Speaker #2: General Electrical Failure	- For a R.F. door speaker concern, GO to Pinpoint Test H - For a R.F. tweeter speaker concern, GO to Pinpoint Test I
B1A02:11	Speaker #2: Circuit Short to Ground	- For a R.F. door speaker concern, GO to Pinpoint Test H - For a R.F. tweeter speaker concern, GO to Pinpoint Test I
B1A02:12	Speaker #2: Circuit Short to Battery	- For a R.F. door speaker concern, GO to Pinpoint Test H - For a R.F. tweeter speaker concern, GO to Pinpoint Test I
B1A02:13	Speaker #2: Circuit Open	- For a R.F. door speaker concern, GO to Pinpoint Test H - For a R.F. tweeter speaker concern, GO to Pinpoint Test I
B1A03:01	Speaker #3: General Electrical Failure	GO to Pinpoint Test J.
B1A03:11	Speaker #3: Circuit Short to Ground	GO to Pinpoint Test J.
B1A03:12	Speaker #3: Circuit Short to Battery	GO to Pinpoint Test J.
B1A03:13	Speaker #3: Circuit Open	GO to Pinpoint Test J.
B1A04:01	Speaker #4: General Electrical Failure	GO to Pinpoint Test K.
B1A04:11	Speaker #4: Circuit Short to Ground	GO to Pinpoint Test K.
B1A04:12	Speaker #4: Circuit Short to Battery	GO to Pinpoint Test K.
B1A04:13	Speaker #4: Circuit Open	GO to Pinpoint Test K.

DTC	Description	Action
B1A56:21	Antenna: Signal Amplitude < Minimum	If DTC B1A56:21 sets after running the bezel diagnostic tests, disregard this DTC. This test is used for end of line or repair bay testing at the Ford plant. Each plant has its own assigned frequency and signal strength requirements. The running and passing of this test depends on local radio stations. If no local station is broadcasting on the required frequency, the test will fail. Therefore, tests ran from the bezel diagnostic screen and the DTC generated from the test can be disregarded.
B1A89:11	Satellite Antenna: Circuit Short To Ground	GO to Pinpoint Test B.
B1A89:13	Satellite Antenna: Circuit Open	GO to Pinpoint Test B.
B1D19:49	Compact Disc Unit: Internal Electronic Failure	• CLEAR the DTC. • WAIT at least 30 seconds. • REPEAT the self-test. • If DTC B1D19:49 is retrieved again, INSTALL a new ACM . REFER to Audio Control Module (ACM).
B1D19:4B	Compact Disc Unit: Over Temperature	The ACM was over temperature. This may occur during extended use. Audio operation can resume after the ACM cools. CLEAR the DTC. This is normal operation.
U0155:00	Lost Communication With Instrument Panel Cluster (IPC) Control Module: No Sub Type Information	GO to Pinpoint Test AA.
U0253:00	Lost Communication With Accessory Protocol Interface Module: No Sub Type Information	GO to Pinpoint Test AC.
U0256:00	Lost Communication With Front Controls Interface Module "A": No Sub Type Information	GO to Pinpoint Test AD.
U0257:00	Lost Communication With Front Controls / Display Interface Module: No Sub Type Information	GO to Pinpoint Test AE.
U201A:51	Control Module Main Calibration Data: Not Programmed	CHECK vehicle service history for recent service actions related to this module. This DTC sets due to incomplete or incorrect PMI procedures. If there have been recent service actions with this module, REPEAT/PERFORM the PMI procedure as directed by the diagnostic scan tool. For more information, REFER to Programmable Module Installation (PMI) in Section 418-01. If there have been no recent service actions, INSTALL a new ACM to correct the failure to retain configuration data. REFER to Audio Control Module (ACM).
U2100:00	Initial Configuration Not Complete: No Sub Type Information	CHECK vehicle service history for recent service actions related to this module. This DTC sets due to incomplete or incorrect PMI procedures. If there have been recent service actions with this module, REPEAT/PERFORM the PMI procedure as directed by the diagnostic scan tool. For more information, REFER to Programmable Module Installation (PMI) in Section 418-01. If there have been no recent service actions, INSTALL a new ACM to correct the failure to retain configuration data. REFER to Audio Control Module (ACM).

DTC	Description	Action
U2101:00	Control Module Configuration Incompatible: No Sub Type Information	CHECK vehicle service history for recent service actions related to this module. This DTC sets due to incomplete or incorrect PMI procedures. If there have been recent service actions with this module, REPEAT/PERFORM the PMI procedure as directed by the diagnostic scan tool. For more information, REFER to Programmable Module Installation (PMI) in Section 418-01. If there have been no recent service actions, INSTALL a new ACM to correct the failure to retain configuration data. REFER to Audio Control Module (ACM).
U3000:41	Control Module: General Checksum Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U3000:41 is retrieved again, INSTALL a new ACM . REFER to Audio Control Module (ACM).
U3000:42	Control Module: General Memory Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U3000:42 is retrieved again, INSTALL a new ACM . REFER to Audio Control Module (ACM).
U3000:96	Control Module: Component Internal Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U3000:96 is retrieved again, INSTALL a new ACM . REFER to Audio Control Module (ACM).
U3003:16	Battery Voltage: Circuit Voltage Below Threshold	GO to Pinpoint Test AF.
U3003:17	Battery Voltage: Circuit Voltage Above Threshold	GO to Pinpoint Test AG.

Accessory Protocol Interface Module (APIM) DTC Chart

DTC	Description	Action
B1252:04	USB Port: System Internal Failure	GO to Pinpoint Test R.
B1252:11	USB Port: Circuit Short to Ground	GO to Pinpoint Test R.
B12B8:04	USB Port #2: System Internal Failure	NOTE: This DTC refers to USB port #2 on the APIM itself, which is not used. • INSPECT the APIM USB port #2 for damage and REMOVE any debris. • CLEAR the DTCs. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC B12B8:04 is retrieved again, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.

DTC	Description	Action
B12B8:11	USB Port #2: Circuit Short To Ground	NOTE: <i>This DTC refers to <u>USB</u> port #2 on the <u>APIM</u> itself, which is not used.</i> • INSPECT the <u>APIM</u> <u>USB</u> port #2 for damage and REMOVE any debris. • CLEAR the DTCs. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC B12B8:11 is retrieved again, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
B1D79:01	Microphone Input: General Electrical Failure	GO to Pinpoint Test P.
B1D79:11	Microphone Input: Circuit Short To Ground	GO to Pinpoint Test P.
B1D79:12	Microphone Input: Circuit Short To Battery	GO to Pinpoint Test P.
B1D79:13	Microphone Input: Circuit Open	GO to Pinpoint Test P.
U0140:00	Lost Communication With Body Control Module: No Sub Type Information	GO to Pinpoint Test Z.
U0155:00	Lost Communication With Instrument Panel Cluster (IPC) Control Module: No Sub Type Information	GO to Pinpoint Test AA.
U0257:00	Lost Communication With Front Controls / Display Interface Module: No Sub Type Information	GO to Pinpoint Test AE.
U0423:00	Invalid Data Received from Instrument Panel Cluster Control Module: No Sub Type Information	RETRIEVE and FOLLOW DTCs present in the <u>IPC</u> . REFER to Section 413-01.
U2100:00	Initial Configuration Not Complete: No Sub Type Information	This <u>DTC</u> sets due to incomplete or incorrect <u>APIM</u> programming/provisioning. CHECK vehicle service history for recent service actions related to the <u>APIM</u> . If there have been recent service actions with the <u>APIM</u> , REPEAT/PERFORM the <u>APIM</u> programming as directed by the scan tool. REFER to <u>APIM</u> Programming in the Description and Operation, Section 415-00C. If there have been no recent service actions, the <u>APIM</u> has failed to retain configuration data. PERFORM the <u>APIM</u> Hardware Test. INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
U2101:00	Control Module Configuration Incompatible: No Sub Type Information	This <u>DTC</u> sets due to incomplete or incorrect <u>APIM</u> programming/provisioning. CHECK vehicle service history for recent service actions related to the <u>APIM</u> . If there have been recent service actions with the <u>APIM</u> , REPEAT/PERFORM the <u>APIM</u> programming as directed by the scan tool. REFER to <u>APIM</u> Programming in the Description and Operation, Section 415-00C. If there have been no recent service actions, the <u>APIM</u> has failed to retain configuration data. PERFORM the <u>APIM</u> Hardware Test. INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
U3000:04	Control Module: System Internal Failure	GO to Pinpoint Test M.

DTC	Description	Action
U3000:41	Control Module: General Checksum Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U3000:41 is retrieved again, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
U3000:42	Control Module: General Memory Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U3000:42 is retrieved again, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
U3000:88	Control Module: Bus Off	NOTE: <i>Network communication/message DTCs can result from intermittent concerns such as damaged wiring or low battery voltage occurrences. Additionally, vehicle repair procedures such as module reprogramming often set these DTCs. Replacing a module to resolve a network DTC is unlikely to resolve the concern. The module was unable to communicate on the network at a point in time. The fault is not currently present.</i> To avoid repeat network concerns: • INSPECT for damaged or corroded wiring at the APIM . • TEST the vehicle battery. REFER to Charging Systems in Section 414-00. • CLEAR the DTC. • REPEAT the network test with the diagnostic scan tool.
U3003:16	Battery Voltage: Circuit Voltage Below Threshold	GO to Pinpoint Test AF.
U3003:17	Battery Voltage: Circuit Voltage Above Threshold	GO to Pinpoint Test AG.

FCDIM DTC Chart

DTC	Description	Action
U0155:00	Lost Communication With Instrument Panel Cluster (IPC) Control Module: No Sub Type Information	GO to Pinpoint Test AA.
U0184:00	Lost Communication With Radio: No Sub Type Information	GO to Pinpoint Test AB.
U0253:00	Lost Communication With Accessory Protocol Interface Module: No Sub Type Information	GO to Pinpoint Test AC.
U0256:00	Lost Communication With Front Controls Interface	GO to Pinpoint Test AD.

DTC	Description	Action
	Module "A": No Sub Type Information	
U201A:51	Control Module Main Calibration Data: Not Programmed	CHECK vehicle service history for recent service actions related to this module. This DTC sets due to incomplete or incorrect PMI procedures. If there have been recent service actions with this module, REPEAT/PERFORM the PMI procedure as directed by the diagnostic scan tool. For more information, REFER to Programmable Module Installation (PMI) in Section 418-01. If there have been no recent service actions, INSTALL a new FCDIM to correct the failure to retain configuration data. REFER to Front Controls/Display Interface Module (FCDIM).
U2100:00	Initial Configuration Not Complete: No Sub Type Information	CHECK vehicle service history for recent service actions related to this module. This DTC sets due to incomplete or incorrect PMI procedures. If there have been recent service actions with this module, REPEAT/PERFORM the PMI procedure as directed by the diagnostic scan tool. For more information, REFER to Programmable Module Installation (PMI) in Section 418-01. If there have been no recent service actions, INSTALL a new FCDIM to correct the failure to retain configuration data. REFER to Front Controls/Display Interface Module (FCDIM).
U2101:00	Control Module Configuration Incompatible: No Sub Type Information	CHECK vehicle service history for recent service actions related to this module. This DTC sets due to incomplete or incorrect PMI procedures. If there have been recent service actions with this module, REPEAT/PERFORM the PMI procedure as directed by the diagnostic scan tool. For more information, REFER to Programmable Module Installation (PMI) in Section 418-01. If there have been no recent service actions, INSTALL a new FCDIM to correct the failure to retain configuration data. REFER to Front Controls/Display Interface Module (FCDIM).
U3000:41	Control Module: General Checksum Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U3000:41 is retrieved again, INSTALL a new FCDIM . REFER to Front Controls/Display Interface Module (FCDIM).
U3000:42	Control Module: General Memory Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U3000:42 is retrieved again, INSTALL a new FCDIM . REFER to Front Controls/Display Interface Module (FCDIM).
U3003:16	Battery Voltage: Circuit Voltage Below Threshold	GO to Pinpoint Test AF.
U3003:17	Battery Voltage: Circuit Voltage Above Threshold	GO to Pinpoint Test AG.

DTC	Description	Action
B00D5:01	Restraint System Passenger Disable Indicator: General Electrical Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC B00D5:01 is retrieved again, INSTALL a new FCIM . REFER to Front Controls Interface Module (FCIM).
B113C:11	Hazard switch illumination: Circuit Short To Ground	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC B113C:11 is retrieved again, INSTALL a new FCIM . REFER to Front Controls Interface Module (FCIM).
B1202:01	Restraint System Passenger Enable Indicator: General Electrical Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC B1202:01 is retrieved again, INSTALL a new FCIM . REFER to Front Controls Interface Module (FCIM).
U0155:00	Lost Communication With Instrument Panel Cluster (IPC) Control Module: No Sub Type Information	GO to Pinpoint Test AA.
U0257:00	Lost Communication With Front Controls / Display Interface Module: No Sub Type Information	GO to Pinpoint Test AE.
U0423:00	Invalid Data Received from Instrument Panel Cluster Control Module: No Sub Type Information	RETRIEVE and FOLLOW DTCs present in the IPC . REFER to Section 413-01.
U2013:63	Switch Pack: Circuit / Component Protection Time-Out	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U2013:63 is retrieved again, INSTALL a new FCIM . REFER to Front Controls Interface Module (FCIM).
U3000:41	Control Module: General Checksum Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U3000:41 is retrieved again, INSTALL a new FCIM . REFER to Front Controls Interface Module (FCIM).
U3000:42	Control Module: General Memory Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U3000:42 is retrieved again, INSTALL a new FCIM . REFER to Front Controls Interface Module (FCIM).
U3003:16	Battery Voltage: Circuit Voltage Below Threshold	GO to Pinpoint Test AF.
U3003:17	Battery Voltage: Circuit Voltage Above Threshold	GO to Pinpoint Test AG.

Global Positioning System Module (GPSM) DTC Chart

DTC	Description	Action
U0100:00	Lost Communication with ECM/PCM "A": No Sub Type Information	GO to Pinpoint Test Y.
U0140:00	Lost Communication With Body Control Module: No Sub Type Information: No Sub Type Information	GO to Pinpoint Test Z.
U0155:00	Lost Communication With Instrument Panel Cluster (IPC) Control Module: No Sub Type Information	GO to Pinpoint Test AA.
U0401:00	Invalid Data Received from ECM/PCM A: No Sub Type Information	RETRIEVE and FOLLOW DTCs present in the PCM . REFER to Section 303-14.
U0422:00	Invalid Data Received From Body Control Module: No Sub Type Information	RETRIEVE and FOLLOW DTCs present in the BCM . REFER to Section 419-10.
U0423:00	Invalid Data Received from Instrument Panel Cluster Control Module: No Sub Type Information	RETRIEVE and FOLLOW DTCs present in the IPC . REFER to Section 413-01.
U2100:00	Initial Configuration Not Complete: No Sub Type Information	CHECK vehicle service history for recent service actions related to this module. This DTC sets due to incomplete or incorrect PMI procedures. If there have been recent service actions with this module, REPEAT/PERFORM the PMI procedure as directed by the diagnostic scan tool. For more information, REFER to Programmable Module Installation (PMI) in Section 418-01. If there have been no recent service actions, INSTALL a new Global Positioning System Module (GPSM) to correct the failure to retain configuration data. REFER to Global Positioning System Module (GPSM).
U2101:00	Control Module Configuration Incompatible: No Sub Type Information	CHECK vehicle service history for recent service actions related to this module. This DTC sets due to incomplete or incorrect PMI procedures. If there have been recent service actions with this module, REPEAT/PERFORM the PMI procedure as directed by the diagnostic scan tool. For more information, REFER to Programmable Module Installation (PMI) in Section 418-01. If there have been no recent service actions, INSTALL a new Global Positioning System Module (GPSM) to correct the failure to retain configuration data. REFER to Global Positioning System Module (GPSM).
U3000:09	Control Module: Component Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U3000:09 is retrieved again, PERFORM a battery reset by disconnecting the battery for 1 minute, then reconnecting it. • CLEAR the DTC.

DTC	Description	Action
		• WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U3000:09 is retrieved again, INSTALL a new Global Positioning System Module (GPSM) . REFER to Global Positioning System Module (GPSM).
U3000:41	Control Module: General Checksum Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U3000:41 is retrieved again, INSTALL a new Global Positioning System Module (GPSM) . REFER to Global Positioning System Module (GPSM).
U3000:42	Control Module: General Memory Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U3000:42 is retrieved again, INSTALL a new Global Positioning System Module (GPSM) . REFER to Global Positioning System Module (GPSM).
U3003:16	Battery Voltage: Circuit Voltage Below Threshold	GO to Pinpoint Test AF.
U3003:17	Battery Voltage: Circuit Voltage Above Threshold	GO to Pinpoint Test AG.

SCCM DTC Chart

DTC	Description	Action
B1380:09	Steering Wheel Right Switch Pack: Component Failure	GO to Pinpoint Test T.
B1380:11	Steering Wheel Right Switch Pack: Circuit Short To Ground	GO to Pinpoint Test T.
B1380:17	Steering Wheel Right Switch Pack: Circuit Voltage Above Threshold	GO to Pinpoint Test T.
All other DTCs	—	REFER to Section 211-05.

Symptom Charts

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Symptom Chart — General Audio System

Condition	Possible Causes	Action
• A module does not respond to the diagnostic scan tool	• Fuse • Wiring, terminals, or connectors	• REFER to Section 418-00.

	Module	
The RH steering wheel controls are inoperative or do not operate correctly	Refer to the Pinpoint Test	GO to Pinpoint Test T.
The CD player is inoperative or does not operate correctly	- CD - ACM	INSPECT the CD for scratches, fingerprints, a loose paper label, incorrect format, or damage. INSERT a known good CD and TEST the system. If the system operates correctly, the concern was caused by a damaged CD. If the system does not operate correctly, INSTALL a new ACM . REFER to Audio Control Module (ACM).
The speed compensated volume does not operate correctly	Refer to the Pinpoint Test	GO to Pinpoint Test C.
The compass is inoperative	Refer to the Pinpoint Test	GO to Pinpoint Test X.
The FCIM illumination is inoperative or does not dim	- Illumination system concern - FCIM	REFER to Section 413-00.
The audio system does not operate correctly from the FCIM	Refer to the Pinpoint Test	GO to Pinpoint Test U.
The FCDIM is inoperative or does not operate correctly	Refer to the Pinpoint Test	GO to Pinpoint Test V.
One or more FCDIM displays are inoperative or do not operate correctly	Refer to the Pinpoint Test	GO to Pinpoint Test W.
The FCDIM illumination is inoperative or does not dim	- Illumination system concern - FCDIM	REFER to Section 413-00.

Symptom Chart — Satellite Radio (If Equipped)

Condition	Possible Causes	Action
Poor reception — satellite radio	Refer to the Pinpoint Test	GO to Pinpoint Test B.

Poor sound quality or no sound while in satellite radio mode - all other functions operate correctly	Refer to the Pinpoint Test	GO to Pinpoint Test B.
The satellite radio is inoperative or does not operate correctly	Refer to the Pinpoint Test	GO to Pinpoint Test B.

Symptom Chart — Sound Quality

Condition	Possible Causes	Action
Poor reception — AM/FM	Refer to the Pinpoint Test	GO to Pinpoint Test A.
Continuous seek or scan — AM/FM	Refer to the Pinpoint Test	GO to Pinpoint Test A.
Poor sound quality or distorted sound from one or more speakers (not all speakers)	Refer to the Pinpoint Test	GO to Pinpoint Test D.
No sound from all speakers	Refer to the Pinpoint Test	GO to Pinpoint Test E.
The LF door speaker is inoperative	Refer to the Pinpoint Test	GO to Pinpoint Test F.
The LF tweeter speaker is inoperative	Refer to the Pinpoint Test	GO to Pinpoint Test G.
The RF door speaker is inoperative	Refer to the Pinpoint Test	GO to Pinpoint Test H.
The RF tweeter speaker is inoperative	Refer to the Pinpoint Test	GO to Pinpoint Test I.
The RR door speaker is inoperative	Refer to the Pinpoint Test	GO to Pinpoint Test J.
The LR door speaker is inoperative	Refer to the Pinpoint Test	GO to Pinpoint Test K.

SYNC System

Condition	Possible Causes	Action
The SYNC® system is inoperative (no response is received from phone, voice, and media inputs)	Refer to the Pinpoint Test	PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test M.
All SYNC® Services features are inoperative or inaccurate (traffic, directions, and information)	Refer to the Pinpoint Test	PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test O.

The SYNC® system voice or tone prompts, TTS feature, or ringtones are inoperative or do not operate correctly	Refer to the Pinpoint Test	PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test N.
During a phone call, no incoming audio is heard in the vehicle	Refer to the Pinpoint Test	PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test N.
Voice recognition is inoperative or does not operate correctly	Refer to the Pinpoint Test	PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test P.
During a phone call, poor quality or no outgoing audio is heard on the outside device	Refer to the Pinpoint Test	PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test P.
No sound from all SYNC® audio sources (Bluetooth, USB , audio input jack)	Refer to the Pinpoint Test	PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test L.
The SYNC® steering wheel switches are inoperative or do not operate correctly	Refer to the Pinpoint Test	PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test T.
The audio input jack is inoperative or does not operate correctly	Refer to the Pinpoint Test	PERFORM the Accessory Protocol Interface Module (APIM) Reset. REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test Q.
The Universal Serial Bus (USB) port is inoperative or does not operate correctly	Refer to the Pinpoint Test	PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test R.
Unable to pair Bluetooth device	Refer to the Pinpoint Test	PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test S.
An individual Bluetooth device feature is inoperative (text messaging, media playback or controls, phone book download,	Customer's device compatibility	- PERFORM the Accessory Protocol Interface Module (APIM) Reset. REFER to Accessory Protocol Interface Module (APIM) Reset. - If the concern is still present, INSTRUCT the customer to review the device compatibility list on the SyncMyRide website. It is normal operation for unsupported or incompatible

call history, or call waiting)		Bluetooth features to be unavailable through the SYNC® system. If the inoperative feature is compatible, ADVISE the customer to check their device for firmware updates, and if required, reset their device following the manufacturer's instructions.
SYNC® AppLink™ is inoperative or does not operate correctly	Refer to the Pinpoint Test	- PERFORM the Accessory Protocol Interface Module (APIM) Reset. REFER to Accessory Protocol Interface Module (APIM) Reset. - If the concern is still present, INSTRUCT the customer to review the mobile device platform, application, and application version compatibility list on the SyncMyRide website. It is normal operation for incompatible mobile device platforms, applications, and application versions to be unavailable through SYNC® AppLink™. - If the mobile device platform, application, and application version are compatible, and the device is connected through Bluetooth only, GO to Pinpoint Test S. - If the mobile device platform, application, and application version are compatible, and the device is connected through Bluetooth and the USB port, GO to Pinpoint Test R
911 Assist™ or Vehicle Health Report (VHR) is inoperative or does not operate correctly	- Unregistered SYNC® owner account - RCM misconfiguration - IPC misconfiguration	- Verify that the customer has a SYNC® Owner Account on the SyncMyRide website. - This error message appears due to incomplete or incorrect PMI procedures. CHECK the vehicle service history for recent service actions related to the IPC or RCM. - If there have been recent service actions with the IPC or RCM, REPEAT/PERFORM the PMI procedure as directed by the diagnostic scan tool. REFER to Programmable Module Installation (PMI) in Section 418-01. - If there have been no recent service actions, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.

Pinpoint Tests

Pinpoint Test A: Poor Reception Or Continuous Seek Or Scan — AM/FM

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to AM/FM Radio, **ACM** , and Antenna in Information and Entertainment System.

The **RDS** function allows the vehicle operator to choose which genre (category) of music to listen to. When the **SCAN** button is pressed, the **ACM** only stops at channels falling within the selected category. If no channel in the selected category is found, the **ACM** continues to scan without stopping. This can appear to be a reception issue, but it is actually due to the customer's setting.

Possible Sources

- AM/FM antenna
- AM/FM antenna coaxial cable
- Charging system
- Ignition system
- No channel found in the selected category
- Noise suppression equipment
- Wiring, terminals, or connectors
- **ACM**

PINPOINT TEST A : POOR RECEPTION OR CONTINUOUS SEEK OR SCAN — AM/FM	
A1 CHECK THE AUDIO SYSTEM RECEPTION	
• Operate the audio system in radio tuner AM/FM mode. • Check the reception with the engine running, and with the engine off.	
Does the poor reception only occur with the engine running?	
Yes	GO to A2 .
No	GO to A4 .
A2 CHECK THE GENERATOR	
• Ignition OFF. • Disconnect: Wiring harness to the generator voltage regulator. • Start the engine. • Operate the audio system in radio tuner AM/FM mode.	
Is the reception OK?	
Yes	INSTALL a new generator. REFER to Section 414-00.
No	GO to A3 .

A3 CHECK THE IGNITION CIRCUITS

- Ignition OFF.
- Connect: Wiring harness to the generator voltage regulator.
- Visually inspect the engine compartment and make sure all ignition coils are correctly and securely connected, and that there are no visible cracks in the coil housings.
- Inspect all the wiring harnesses and connectors for damaged insulation and loose or broken conditions.

Are the ignition components OK?

Yes	USE a jumper cable to ground various parts of the vehicle to the frame (for example: engine, fenders, quarter panels, stone deflectors, air cleaner, body sheet metal). When the noise is eliminated, PROVIDE a permanent ground where necessary.
No	REPAIR the ignition system as necessary.

A4 CHECK THE LOWER AM/FM ANTENNA COAXIAL CABLE FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: AM/FM antenna coaxial cable at the ACM . Refer to Audio Control Module (ACM).
- Disconnect: AM/FM antenna inline coaxial cable connection behind the glove compartment. Refer to Antenna Cable — AM/FM.
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the <u>ACM</u>	—	—	Ground
AM/FM antenna coaxial cable shield at the <u>ACM</u>	—	—	Ground

Is any voltage present?

Yes	REPAIR or INSTALL a new lower AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM.
No	GO to A5 .

A5 CHECK THE LOWER AM/FM ANTENNA COAXIAL CABLE CORE FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the <u>ACM</u>		—	Ground

Is the resistance greater than 10,000 ohms?

Yes	GO to A6 .
No	REPAIR or INSTALL a new lower AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM.

A6 CHECK THE LOWER AM/FM ANTENNA COAXIAL CABLE FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM coaxial cable core at the <u>ACM</u>	—	Lower AM/FM antenna coaxial cable core at the inline connection	—
AM/FM antenna coaxial cable shield at the <u>ACM</u>	—	Lower AM/FM antenna coaxial cable shield at the inline connection	—

Is the resistance less than 1 ohm?

Yes	GO to A7 .
No	REPAIR or INSTALL a new lower AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM.

A7 CHECK THE LOWER AM/FM ANTENNA COAXIAL CABLE CORE AND SHIELD FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the <u>ACM</u>	—	AM/FM antenna coaxial cable shield at the <u>ACM</u>	—

Is the resistance greater than 10,000 ohms?

Yes	GO to A8 .
No	REPAIR or INSTALL a new lower AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM.

A8 CHECK THE UPPER AM/FM ANTENNA COAXIAL CABLE FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Connect: AM/FM antenna inline coaxial cable connection behind the glove compartment.
- Disconnect: AM/FM antenna coaxial cable at the antenna unit. Refer to Antenna — AM/FM.
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the ACM	—	—	Ground
AM/FM antenna coaxial cable shield at the ACM	—	—	Ground

Is any voltage present?

Yes	REPAIR or INSTALL a new upper AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM.
No	GO to A9 .

A9 CHECK THE UPPER AM/FM ANTENNA COAXIAL CABLE CORE FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the ACM	—	—	Ground

Is the resistance greater than 10,000 ohms?

Yes	GO to A10 .
No	REPAIR or INSTALL a new upper AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM.

A10 CHECK THE UPPER AM/FM ANTENNA COAXIAL CABLE FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the ACM	—	AM/FM antenna coaxial cable core at the antenna unit (harness side)	—
AM/FM antenna coaxial cable shield at the ACM	—	AM/FM antenna coaxial cable shield at the antenna unit (harness side)	—

Is the resistance less than 1 ohm?

Yes	GO to A11 .
No	REPAIR or INSTALL a new upper AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM.

A11 CHECK THE UPPER AM/FM ANTENNA COAXIAL CABLE CORE AND SHIELD FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the ACM	—	AM/FM antenna coaxial cable shield at the ACM	—

Is the resistance greater than 10,000 ohms?

Yes	GO to A12 .
No	REPAIR or INSTALL a new upper AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM.

A12 CHECK THE AM/FM ANTENNA UNIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Connect: AM/FM antenna coaxial cable at the antenna unit.
- Ignition ON.

- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the <u>ACM</u>	—	—	Ground
AM/FM antenna coaxial cable shield at the <u>ACM</u>	—	—	Ground

Is any voltage present?

Yes	INSTALL a new AM/FM antenna. REFER to Antenna — AM/FM.
No	GO to A13 .

A13 CHECK THE AM/FM ANTENNA UNIT CORE FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the <u>ACM</u>	—	—	Ground

Is the resistance greater than 10,000 ohms?

Yes	GO to A14 .
No	INSTALL a new AM/FM antenna. REFER to Antenna — AM/FM.

A14 CHECK THE AM/FM ANTENNA UNIT CORE AND SHIELD FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the <u>ACM</u>	—	AM/FM antenna coaxial cable shield at the <u>ACM</u>	—

Is the resistance greater than 10,000 ohms?

Yes	GO to A15 .
No	INSTALL a new AM/FM antenna. REFER to Antenna — AM/FM.

A15 CHECK THE AM/FM ANTENNA UNIT SHIELD FOR CONTINUITY TO GROUND

- Disconnect: AM/FM antenna coaxial cable at the antenna unit.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna unit shield at the antenna unit (component side)	—	—	Ground

Is the resistance less than 1 ohm?

Yes	GO to A17 .
No	GO to A16 .

A16 CHECK THE AM/FM ANTENNA UNIT SHIELD FOR CONTINUITY TO ANTENNA UNIT BASE

- Remove the AM/FM antenna. Refer to Antenna — AM/FM.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna unit shield at the antenna unit (component side)	—	AM/FM antenna base (unpainted surface)	—

Is the resistance less than 1 ohm?

Yes	GO to A17 .
No	INSTALL a new AM/FM antenna. REFER to Antenna — AM/FM.

A17 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the ~~ACM~~ connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)

- Damaged or bent pins (install new terminals or pins)
- Pushed-out pins (install new pins as necessary)

- Reconnect the **ACM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test B: The Satellite Radio Is Inoperative, Has Poor Reception, Or Has No Sound

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to MyKey® and Satellite Radio in Information and Entertainment System.

When a satellite radio subscription is activated, the satellite receiver **ESN** is associated with the **VIN** .

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
B1A89:11	Satellite Antenna: Circuit Short To Ground	Set by the ACM when a short to ground is detected in the satellite antenna circuit for more than 250 milliseconds. This DTC can be either continuous or on-demand.
B1A89:13	Satellite Antenna: Circuit Open	Set by the ACM when an open is detected in the satellite antenna circuit. This DTC can be either continuous or on-demand.

Possible Sources

- Expired subscription
- Incorrectly matched **ESN**
- Wiring, terminals, or connectors
- Obstructions in the antenna's line of sight
- Satellite radio antenna
- Satellite radio antenna coaxial cable

- ACM

PINPOINT TEST B : THE SATELLITE RADIO IS INOPERATIVE, HAS POOR RECEPTION, OR HAS NO SOUND

B1 VERIFY AN ACTIVE SUBSCRIPTION

- Operate the infotainment system in satellite radio mode and observe the infotainment display.

Does the display indicate that the subscription has expired?

Yes	The subscription has expired. INFORM the customer to contact SIRIUS to reactivate the subscription.
No	GO to B2 .

B2 CHECK THE RECEPTION

- Drive the vehicle to an open location that is free from obstacles such as tall buildings or large trees.
- Using a diagnostic scan tool, view the ~~ACM~~ PIDs.
- Monitor the satellite signal strength (SAT_SIG_STR) PID.
- If the PID reads "No Signal" or "Poor," move the vehicle to another area and retest until "Good," "Very Good," or "Excellent" PID values are obtained.

Can a "Good," "Very Good," or "Excellent" PID value be obtained?

Yes	GO to B3 .
No	GO to B6 .

B3 CHECK THE SATELLITE RADIO SYSTEM FOR CORRECT OPERATION

- Operate multiple channels with the infotainment system in satellite radio mode.

Does the satellite radio system receive multiple channels with normal sound quality?

Yes	The system is operating correctly at this time. The concern was a result of obstructions in the antenna's line of sight.
No	GO to B4 .

B4 CHECK FOR PROMOTIONAL CHANNEL 184

- Select channel 184.

Does the satellite radio system receive channel 184 with normal sound quality?

Yes	GO to B5 .
No	GO to B6 .

B5 VERIFY THAT THE VIN AND ESN ON FILE MATCH THE VIN AND ESN FROM THE VEHICLE

- Retrieve the satellite radio ESN . Refer to Bezel Diagnostics in this section.
- Contact the SIRIUS Dealer Support Line (888-465-8528).
- Ensure that the VIN and ESN on the customer's file match the VIN and ESN retrieved from the vehicle.
- Follow the actions from step 2 to obtain "Good," "Very Good," or "Excellent" reception.
- Operate the infotainment system in satellite radio mode.
- **NOTE:** *It may take up to 30 minutes for the activation signal to be received and satellite radio functions to operate.*

Request for SIRIUS to send the activation signal to the vehicle 2-3 times.

- Operate multiple channels with the infotainment system in satellite radio mode.

Does the satellite radio system receive multiple channels with normal sound quality?

Yes	The system is operating correctly at this time. The concern was a result of the activation signal needing to be resent or the <u>VIN</u> and <u>ESN</u> SIRIUS had on file not matching the <u>VIN</u> and <u>ESN</u> from the vehicle.
No	GO to B6 .

B6 CHECK THE LOWER SATELLITE RADIO ANTENNA COAXIAL CABLE FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: Satellite radio antenna coaxial cable at the ACM . Refer to Audio Control Module (ACM).
- Disconnect: Satellite radio antenna inline coaxial cable connection behind the glove compartment. Refer to Antenna Cable — Satellite Radio.
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna coaxial cable core at the <u>ACM</u>	—	—	Ground
Satellite radio antenna coaxial cable shield at the <u>ACM</u>	—	—	Ground

Is any voltage present?

Yes	REPAIR or INSTALL a new lower satellite radio antenna coaxial cable. REFER to Antenna Cable
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	— Satellite Radio.
No	GO to B7 .

B7 CHECK THE LOWER SATELLITE RADIO ANTENNA COAXIAL CABLE CORE FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna coaxial cable core at the ACM	—	—	Ground

Is the resistance greater than 10,000 ohms?

Yes	GO to B8 .
No	REPAIR or INSTALL a new lower satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio.

B8 CHECK THE LOWER SATELLITE RADIO ANTENNA COAXIAL CABLE FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio coaxial cable core at the ACM	—	Lower satellite radio antenna coaxial cable core at the inline connection	—
Satellite radio antenna coaxial cable shield at the ACM	—	Lower satellite radio antenna coaxial cable shield at the inline connection	—

Is the resistance less than 1 ohm?

Yes	GO to B9 .
No	REPAIR or INSTALL a new lower satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio.

B9 CHECK THE LOWER SATELLITE RADIO ANTENNA COAXIAL CABLE CORE AND SHIELD FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna coaxial cable core at the ACM	—	Satellite radio antenna coaxial cable shield at the ACM	—

Is the resistance greater than 10,000 ohms?

Yes	GO to B10 .
No	REPAIR or INSTALL a new lower satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio.

B10 CHECK THE UPPER SATELLITE RADIO ANTENNA COAXIAL CABLE FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Connect: Satellite radio antenna inline coaxial cable connection behind the glove compartment.
- Disconnect: Satellite radio antenna coaxial cable at the antenna unit. Refer to Antenna — Satellite Radio.
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna coaxial cable core at the ACM	—	—	Ground
Satellite radio antenna coaxial cable shield at the ACM	—	—	Ground

Is any voltage present?

Yes	REPAIR or INSTALL a new upper satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio.
No	GO to B11 .

B11 CHECK THE UPPER SATELLITE RADIO ANTENNA COAXIAL CABLE CORE FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna coaxial cable core at the ACM	—	—	Ground

Is the resistance greater than 10,000 ohms?

Yes	GO to B12 .
No	REPAIR or INSTALL a new upper satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio.

B12 CHECK THE UPPER SATELLITE RADIO ANTENNA COAXIAL CABLE FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna coaxial cable core at the ACM	—	Satellite radio antenna coaxial cable core at the antenna unit (harness side)	—
Satellite radio antenna coaxial cable shield at the ACM	—	Satellite radio antenna coaxial cable shield at the antenna unit (harness side)	—

Is the resistance less than 1 ohm?

Yes	GO to B13 .
No	REPAIR or INSTALL a new upper satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio.

B13 CHECK THE UPPER SATELLITE RADIO ANTENNA COAXIAL CABLE CORE AND SHIELD FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna coaxial cable core at the ACM	—	Satellite radio antenna coaxial cable shield at the ACM	—

Is the resistance greater than 10,000 ohms?

Yes	GO to B14 .
No	REPAIR or INSTALL a new upper satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio.

B14 CHECK THE SATELLITE RADIO ANTENNA UNIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Connect: Satellite radio antenna coaxial cable at the antenna unit.
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna coaxial cable core at the ACM	—	—	Ground
Satellite radio antenna coaxial cable shield at the ACM	—	—	Ground

Is any voltage present?

Yes	INSTALL a new satellite radio antenna. REFER to Antenna — Satellite Radio.
No	GO to B15 .

B15 CHECK THE SATELLITE RADIO ANTENNA UNIT CORE FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna coaxial cable core at the ACM	—	—	Ground

Is the resistance greater than 10,000 ohms?

Yes	GO to B16 .
No	INSTALL a new satellite radio antenna. REFER to Antenna — Satellite Radio.

B16 CHECK THE SATELLITE RADIO ANTENNA UNIT CORE AND SHIELD FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna coaxial cable core at the ACM	—	Satellite radio antenna coaxial cable shield at the ACM	—

Is the resistance greater than 10,000 ohms?

Yes	GO to B17 .
No	INSTALL a new satellite radio antenna. REFER to Antenna — Satellite Radio.

B17 CHECK THE SATELLITE RADIO ANTENNA UNIT SHIELD FOR CONTINUITY TO GROUND

- Disconnect: Satellite radio antenna coaxial cable at the antenna unit.
- Measure the **resistance** between:

Positive Lead			Negative Lead	
Pin		Circuit	Pin	Circuit
Satellite radio antenna unit shield at the antenna unit (component side)		—	—	Ground

Is the resistance less than 1 ohm?

Yes	GO to B19 .
No	GO to B18 .

B18 CHECK THE SATELLITE RADIO ANTENNA UNIT SHIELD FOR CONTINUITY TO ANTENNA UNIT BASE

- Remove the satellite radio antenna. Refer to Antenna — Satellite Radio.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna unit shield at the antenna unit (component side)	—	Satellite radio antenna base (unpainted surface)	—

Is the resistance less than 1 ohm?

Yes	GO to B19 .
No	INSTALL a new satellite radio antenna. REFER to Antenna — Satellite Radio.

B19 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the ACM connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the ACM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test C: The Speed Compensated Volume Does Not Operate Correctly

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Normal Operation and Fault Conditions

Refer to Speed Compensated Volume in Information and Entertainment System.

Possible Sources

- Speed compensated volume setting
- YSS signal concern
- Network communication concern
- ACM

PINPOINT TEST C : THE SPEED COMPENSATED VOLUME DOES NOT OPERATE CORRECTLY

C1 CHECK THE SPEEDOMETER OPERATION

- Drive the vehicle and observe the speedometer.

Does the speedometer operate correctly?

Yes	GO to C2 .
No	REFER to Section 413-01.

C2 CHECK THE SPEED COMPENSATED VOLUME SETTING

- Turn the speed compensated volume off. Refer to the Owner's Literature.
- Operate the audio system in radio tuner AM/FM mode.
- Drive the vehicle at various speeds and observe the speaker volume.
- Set the speed compensated volume to maximum compensation. Refer to the Owner's Literature.
- Operate the audio system in radio tuner AM/FM mode.
- Drive the vehicle at various speeds and observe the speaker volume.

Does the volume remain constant with the speed compensated volume turned off, and increase and decrease with vehicle speed with the speed compensated volume set to maximum?

Yes	The system is operating correctly at this time. INSTRUCT the customer in the correct usage of the speed compensated volume feature.
No	GO to C3 .

C3 CHECK FOR DTC U0155:00

- Using a diagnostic scan tool, retrieve all CMDTCs .

Is DTC U0155:00 retrieved in any audio system module?

Yes	GO to Pinpoint Test AA .
No	GO to C4 .

C4 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the ACM connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the ACM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test D: Poor Sound Quality Or Distorted Sound (Buzz/Crackle/Rattle) From One Or More Speakers (Not All Speakers)

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Normal Operation and Fault Conditions

The ACM sends audio signals to the speakers in the form of AC voltage, resulting in clear audio output. REFER to Information and Entertainment System. See USB Audio Mode.

Possible Sources

- Loose component fasteners
- Loose door handle, lock, and other trim panel components and fasteners
- Loose items in storage areas
- Loose speaker grilles
- Loose wire harnesses or wire harness fasteners
- Speaker(s)
- Water shield bonding and placement

Visual Inspection and Diagnostic Pre-checks

Visually inspect around the suspect area for any possible rattle conditions such as:

- Any storage compartments including cup holders (if equipped)
- Door map pockets for contents that can rattle (if equipped)
- Door pull cup attachments (if equipped)
- Door switch bezels (if equipped)
- Package tray (if equipped)
- Speaker grilles
- Trim panels and fasteners

PINPOINT TEST D : POOR SOUND QUALITY OR DISTORTED SOUND (BUZZ/CRACKLE/RATTLE) FROM ONE OR MORE SPEAKERS (NOT ALL SPEAKERS)

D1 ISOLATE THE ZONE
• Download the diagnostic sound tracks to a suitable USB drive then install the USB drive into the Media Hub. Refer to Speaker Audio Test. • On the audio system select USB input. • Using the audio system fade and balance feature, adjust the audio sound to each of the four zones (LF , RF , LR , and RR) of the vehicle and use the 6 diagnostic sound tracks to isolate the poor sound quality. Allow each track to run for 10 seconds, then use the Seek buttons to change the test track.

- Locate and apply pressure to the trim panel(s) around the poor sound quality area in question.

Does applying pressure to a trim panel reduce or eliminate the audible noise?

Yes	REPAIR or REPLACE the trim panel as needed.
No	GO to D2 .

D2 REMOVE AND INSPECT BEHIND/UNDERNEATH THE SUSPECT TRIM PANEL(S)

- Remove the trim panel to access the suspect speaker.
- Operate the audio system using digital media (CD, MP3, etc.).
- Validate where the poor sound quality area is located.
- Check:
 - Trim panel for loose components around the speaker area such as the speaker grille
 - Trim panel joining components for missing or broken pieces
 - Lock and handle mechanical parts for correct attachment
 - Wire harnesses for correct routing
 - Wire harness fasteners for correct attachment
 - Water shield for correct placement (speaker air path not blocked)
 - Water shield for correct bonding to sheet metal
 - Storage areas for loose items
 - All child safety belt anchors (if equipped)
 - All safety belt retractors
 - All speaker bracket fasteners and other fasteners are secured and tightened to specified torque

Was the source of the noise located?

Yes	REPAIR or REPLACE any loose or broken component or fastener as needed.
No	GO to D3 .

D3 CHECK THE SUSPECT SPEAKER FOR WATER INTRUSION

- Inspect for watermarks.
- Check the:
 - Cone
 - Magnet
 - Basket

Are any watermarks present on the speaker?

Yes	VERIFY the water shield is in the correct location, REPAIR or REPLACE any trim, door, or speaker seal as required. DRY the speaker in question and TEST the system for normal operation.
No	GO to D4 .

D4 ISOLATE SPEAKER TO VERIFY NOISE

- Remove the suspect speaker from its location and leave the speaker connected to the harness.
- Hold the speaker away from any trim panel(s) and ensure the speaker is isolated from contact.
- Operate the audio system using digital media (CD, MP3, etc.).

Is the noise still present in the suspect speaker?

Yes	INSTALL a new speaker for the one in question.
No	LOCATE the source of the noise and REPAIR as needed.

Pinpoint Test E: No Sound From All Speakers

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to **ACM** in Information and Entertainment System.

When the ignition is in the START position, the **ACM** receives voltage on the START signal circuit and mutes the speaker output. This prevents voltage spikes from producing popping noises through the speakers. A short to voltage on this circuit causes the **ACM** output to be muted continuously.

A short to ground or short to voltage in the circuitry to one of the speakers may cause multiple speakers to lose sound due to the built-in overload protection feature of the **ACM** . In this case, a speaker fault DTC sets.

Possible Sources

- Wiring, terminals, or connectors
- **ACM**

PINPOINT TEST E : NO SOUND FROM ALL SPEAKERS

E1 CHECK FOR SPEAKER DTCS

- Ignition ON.
- Using a diagnostic scan tool, perform the **ACM** self-test.

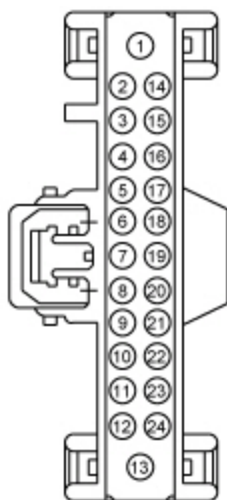
Are any speaker fault Diagnostic Trouble Codes (DTCs) present?

Yes	REFER to the ACM DTC chart in Information and Entertainment System.
No	GO to E2 .

E2 CHECK THE START SENSE CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: ~~ACM~~ [C240A](#) .
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C240A Pin 3	CDC54 (WH/GN)	—	Ground



Is the voltage 5V or greater?

Yes	REPAIR the circuit. TEST the system for normal operation.
No	GO to E3 .

E3 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the ~~ACM~~ connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the ~~ACM~~ connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test F: The **LF** Door Speaker Is Inoperative

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to **ACM** in Information and Entertainment System.

Root DTC Description	Failure Types	Fault Trigger Conditions
B1A01:01	Speaker #1: General Electrical Failure	Set by the ACM when a failure is detected on the LF door or tweeter speaker circuits.
B1A01:11	Speaker #1: Circuit Short to Ground	Set by the ACM when a short to ground is detected on the LF door or tweeter speaker circuits. When DTC B1A01:11 is set, the speaker output is disabled.
B1A01:12	Speaker #1: Circuit Short to Battery	Set by the ACM when a short to battery is detected on the LF door or tweeter speaker circuits. When DTC B1A01:12 is set, the speaker output is disabled.
B1A01:13	Speaker #1: Circuit Open	Set by the ACM when an open is detected on the LF door or tweeter speaker circuits.

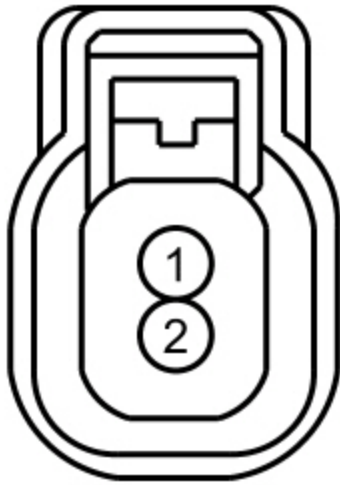
Possible Sources

- Wiring, terminals, or connectors
- Speaker
- **ACM**

PINPOINT TEST F : THE LF DOOR SPEAKER IS INOPERATIVE

F1 CHECK THE AUDIO SIGNAL TO THE LF DOOR SPEAKER
• Ignition OFF. • Disconnect: LF Door Speaker C523. • Operate the audio system in radio tuner AM/FM mode. • Measure the AC voltage between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C523 Pin 1	VME07 (WH)	C523 Pin 2	RME07 (WH/BN)



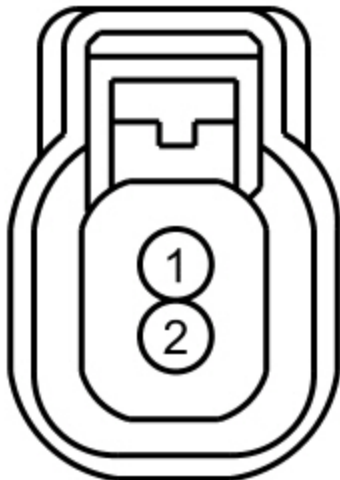
Is a fluctuating AC voltage present?

Yes	INSTALL a new LF door speaker. REFER to Speaker — Door.
No	GO to F2 .

F2 CHECK THE AUDIO CIRCUITS TO THE LF DOOR SPEAKER FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: ACM [C240A](#) .
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C523 Pin 1	VME07 (WH)	—	Ground
C523 Pin 2	RME07 (WH/BN)	—	Ground



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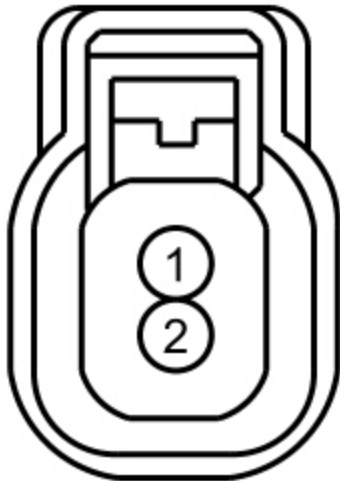
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to F3 .

F3 CHECK THE AUDIO CIRCUITS TO THE LF DOOR SPEAKER FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect: L.F. Tweeter Speaker [C230](#).
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C523 Pin 1	VME07 (WH)	—	Ground
C523 Pin 2	RME07 (WH/BN)	—	Ground



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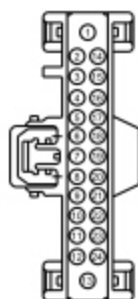
Are the resistances greater than 10,000 ohms?

Yes	GO to F4 .
No	REPAIR the circuit in question.

F4 CHECK THE AUDIO CIRCUITS TO THE LF DOOR SPEAKER FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C523 Pin 1	VME07 (WH)	C240A Pin 22	VME07 (WH)
C523 Pin 2	RME07 (WH/BN)	C240A Pin 10	RME07 (WH/BN)



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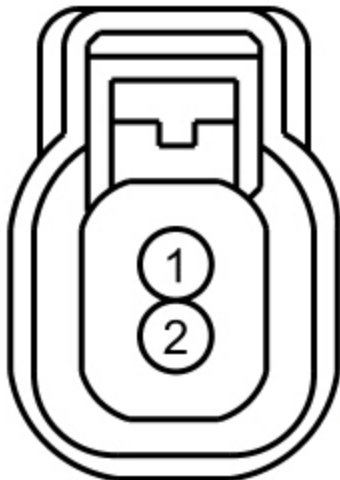
Are the resistances less than 3 ohms?

Yes	GO to F5 .
No	REPAIR the circuit in question.

F5 CHECK THE AUDIO CIRCUITS TO THE LF DOOR SPEAKER FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C523 Pin 1	VME07 (WH)	C523 Pin 2	RME07 (WH/BN)



•

Are the resistances greater than 10,000 ohms?

Yes	GO to F6 .
No	REPAIR the circuit in question.

F6 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the **ACM** connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **ACM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test G: The LRF Tweeter Speaker Is Inoperative

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to [ACM](#) in Information and Entertainment System.

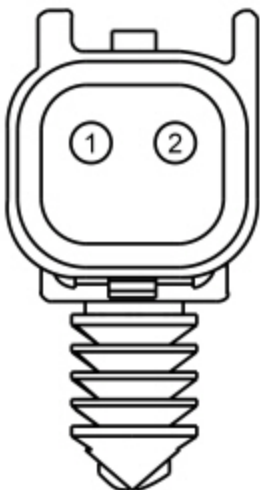
Root DTC Description	Failure Types	Fault Trigger Conditions
B1A01:01	Speaker #1: General Electrical Failure	Set by the ACM when a failure is detected on the LF door or tweeter speaker circuits.
B1A01:11	Speaker #1: Circuit Short to Ground	Set by the ACM when a short to ground is detected on the LF door or tweeter speaker circuits. When DTC B1A01:11 is set, the speaker output is disabled.
B1A01:12	Speaker #1: Circuit Short to Battery	Set by the ACM when a short to battery is detected on the LF door or tweeter speaker circuits. When DTC B1A01:12 is set, the speaker output is disabled.
B1A01:13	Speaker #1: Circuit Open	Set by the ACM when an open is detected on the LF door or tweeter speaker circuits.

Possible Sources

- Wiring, terminals, or connectors
- Speaker
- [ACM](#)

PINPOINT TEST G : THE LF TWEETER SPEAKER IS INOPERATIVE

G1 CHECK THE AUDIO SIGNAL TO THE LF TWEETER SPEAKER			
• Ignition OFF. • Disconnect: LF Tweeter Speaker C230. • Operate the audio system in radio tuner AM/FM mode. • Measure the AC voltage between:			
Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C230 Pin 1	VME07 (WH)	C230 Pin 2	RME07 (WH/BN)



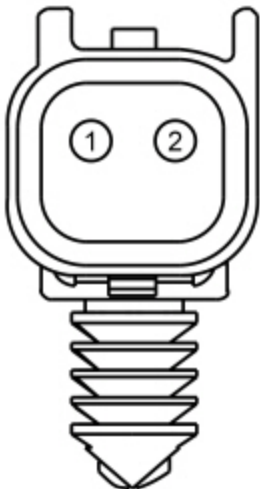
Is a fluctuating AC voltage present?

Yes	INSTALL a new LF tweeter speaker. REFER to Speaker — A-Pillar.
No	GO to G2 .

G2 CHECK THE AUDIO CIRCUITS TO THE LF TWEETER SPEAKER FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: ACM [C240A](#) .
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C230 Pin 1	VME07 (WH)	—	Ground
C230 Pin 2	RME07 (WH/BN)	—	Ground



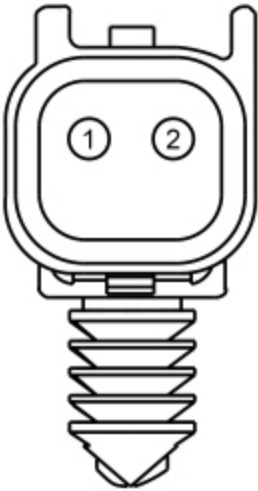
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to G3 .

G3 CHECK THE AUDIO CIRCUITS TO THE LF TWEETER SPEAKER FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect: LF Door Speaker [C523](#).
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C230 Pin 1	VME07 (WH)	—	Ground
C230 Pin 2	RME07 (WH/BN)	—	Ground



Are the resistances greater than 10,000 ohms?

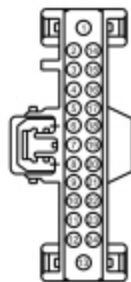
Yes	GO to G4 .
No	REPAIR the circuit in question.

G4 CHECK THE AUDIO CIRCUITS TO THE LF TWEETER SPEAKER FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C230 Pin 1	VME07 (WH)	C240A Pin 22	VME07 (WH)

C230 Pin 2	RME07 (WH/BN)	C240A Pin 10	RME07 (WH/BN)
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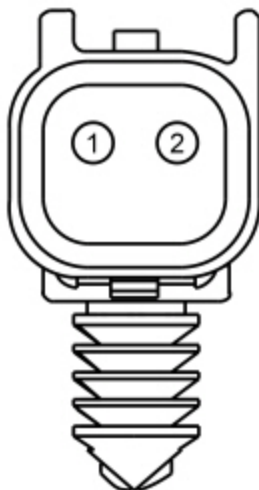
Are the resistances less than 3 ohms?

Yes	GO to G5 .
No	REPAIR the circuit in question.

G5 CHECK THE AUDIO CIRCUITS TO THE LF TWEETER SPEAKER FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C230 Pin 1	VME07 (WH)	C230 Pin 2	RME07 (WH/BN)



Are the resistances greater than 10,000 ohms?

Yes	GO to G6 .
No	REPAIR the circuit in question.

G6 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the **ACM** connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **ACM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test H: The **RF** Door Speaker Is Inoperative

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to **ACM** in Information and Entertainment System.

Root DTC Description	Failure Types	Fault Trigger Conditions
B1A02:01	Speaker #2: General Electrical Failure	Set by the ACM when a failure is detected on the RF door or tweeter speaker circuits.
B1A02:11	Speaker #2: Circuit Short to Ground	Set by the ACM when a short to ground is detected on the RF door or tweeter speaker circuits. When DTC B1A02:11 is set, the speaker output is disabled.
B1A02:12	Speaker #2: Circuit Short to Battery	Set by the ACM when a short to battery is detected on the RF door or tweeter speaker circuits. When DTC B1A02:12 is set, the speaker output is disabled.
B1A02:13	Speaker #2: Circuit Open	Set by the ACM when an open is detected on the RF door or tweeter speaker circuits.

Possible Sources

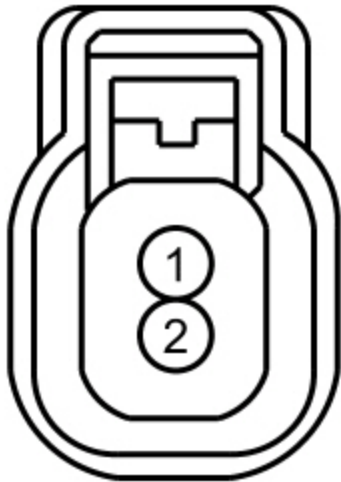
- Wiring, terminals, or connectors
- Speaker
- ACM

PINPOINT TEST H : THE RF DOOR SPEAKER IS INOPERATIVE

H1 CHECK THE AUDIO SIGNAL TO THE RF DOOR SPEAKER

- Ignition OFF.
- Disconnect: RF Door Speaker [C612](#).
- Operate the audio system in radio tuner AM/FM mode.
- Measure the **AC voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C612 Pin 1	VME10 (WH/VT)	C612 Pin 2	RME10 (WH/OG)



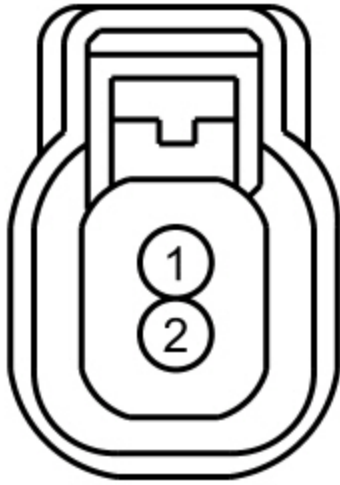
Is a fluctuating AC voltage present?

Yes	INSTALL a new RF door speaker. REFER to Speaker — Door.
No	GO to H2 .

H2 CHECK THE AUDIO CIRCUITS TO THE RF DOOR SPEAKER FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: ACM [C240A](#) .
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C612 Pin 1	VME10 (WH/VT)	—	Ground
C612 Pin 2	RME10 (WH/OG)	—	Ground



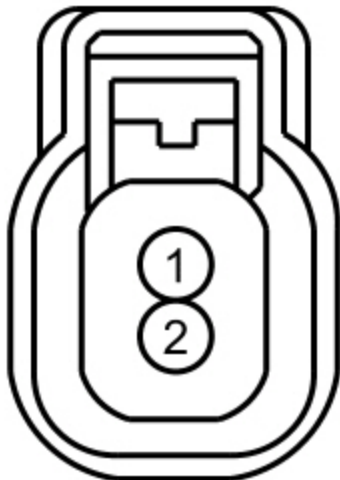
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to H3 .

H3 CHECK THE AUDIO CIRCUITS TO THE RF DOOR SPEAKER FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect: RF Door Speaker [C231](#).
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C612 Pin 1	VME10 (WH/VT)	—	Ground
C612 Pin 2	RME10 (WH/OG)	—	Ground



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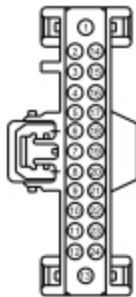
Are the resistances greater than 10,000 ohms?

Yes	GO to H4 .
No	REPAIR the circuit in question.

H4 CHECK THE AUDIO CIRCUITS TO THE RF DOOR SPEAKER FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C612 Pin 1	VME10 (WH/VT)	C240A Pin 23	VME10 (WH/VT)
C612 Pin 2	RME10 (WH/OG)	C240A Pin 11	RME10 (WH/OG)



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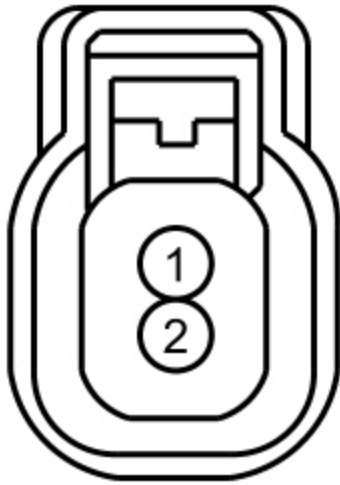
Are the resistances less than 3 ohms?

Yes	GO to H5 .
No	REPAIR the circuit in question.

H5 CHECK THE AUDIO CIRCUITS TO THE RF DOOR SPEAKER FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C612 Pin 1	VME10 (WH/VT)	C612 Pin 2	RME10 (WH/OG)



Are the resistances greater than 10,000 ohms?

Yes	GO to H6 .
No	REPAIR the circuit in question.

H6 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the **ACM** connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **ACM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test I: The RF Tweeter Speaker Is Inoperative

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to [ACM](#) in Information and Entertainment System.

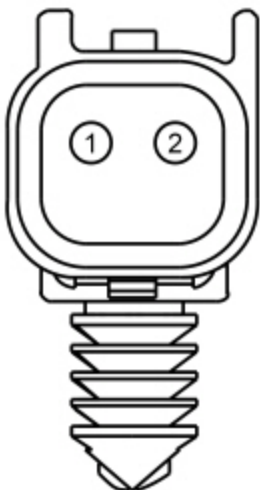
Root DTC Description	Failure Types	Fault Trigger Conditions
B1A02:01	Speaker #2: General Electrical Failure	Set by the ACM when a failure is detected on the RF door or tweeter speaker circuits.
B1A02:11	Speaker #2: Circuit Short to Ground	Set by the ACM when a short to ground is detected on the RF door or tweeter speaker circuits. When DTC B1A02:11 is set, the speaker output is disabled.
B1A02:12	Speaker #2: Circuit Short to Battery	Set by the ACM when a short to battery is detected on the RF door or tweeter speaker circuits. When DTC B1A02:12 is set, the speaker output is disabled.
B1A02:13	Speaker #2: Circuit Open	Set by the ACM when an open is detected on the RF door or tweeter speaker circuits.

Possible Sources

- Wiring, terminals, or connectors
- Speaker
- [ACM](#)

PINPOINT TEST I : THE RF TWEETER SPEAKER IS INOPERATIVE

I1 CHECK THE AUDIO SIGNAL TO THE RF TWEETER SPEAKER			
• Ignition OFF. • Disconnect: RF Tweeter Speaker C231. • Operate the audio system in radio tuner AM/FM mode. • Measure the AC voltage between:			
Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C231 Pin 1	VME10 (WH/VT)	C231 Pin 2	RME10 (WH/OG)



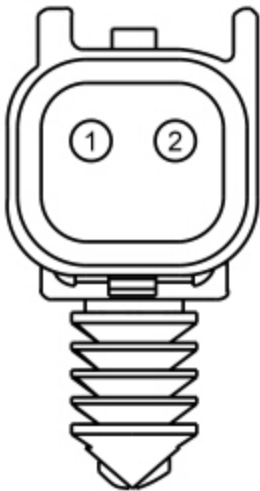
Is a fluctuating AC voltage present?

Yes	INSTALL a new RF tweeter speaker. REFER to Speaker — A-Pillar.
No	GO to I2 .

I2 CHECK THE AUDIO CIRCUITS TO THE RF TWEETER SPEAKER FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: ACM [C240A](#) .
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C231 Pin 1	VME10 (WH/VT)	—	Ground
C231 Pin 2	RME10 (WH/OG)	—	Ground



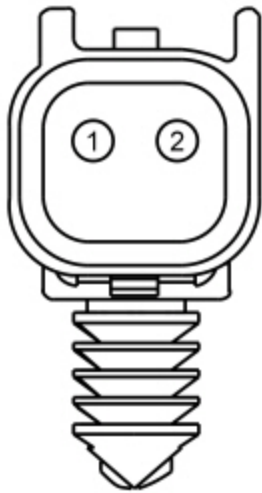
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to I3 .

I3 CHECK THE AUDIO CIRCUITS TO THE RF TWEETER SPEAKER FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect: RF Door Speaker [C612](#).
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C231 Pin 1	VME10 (WH/VT)	—	Ground
C231 Pin 2	RME10 (WH/OG)	—	Ground



Are the resistances greater than 10,000 ohms?

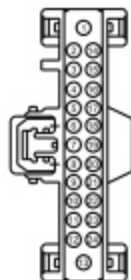
Yes	GO to I4 .
No	REPAIR the circuit in question.

I4 CHECK THE AUDIO CIRCUITS TO THE RF TWEETER SPEAKER FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C231 Pin 1	VME10 (WH/VT)	C240A Pin 23	VME10 (WH/VT)

C231 Pin 2	RME10 (WH/OG)	C240A Pin 11	RME10 (WH/OG)
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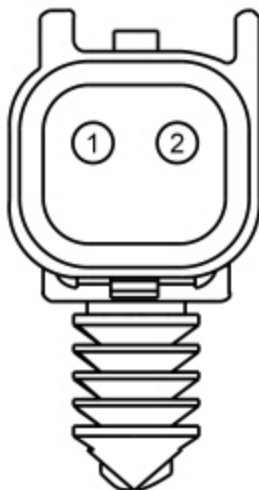
Are the resistances less than 3 ohms?

Yes	GO to I5 .
No	REPAIR the circuit in question.

I5 CHECK THE AUDIO CIRCUITS TO THE RF TWEETER SPEAKER FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C231 Pin 1	VME10 (WH/VT)	C231 Pin 2	RME10 (WH/OG)



Are the resistances greater than 10,000 ohms?

Yes	GO to I6 .
No	REPAIR the circuit in question.

I6 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the ACM connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the ACM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test J: The RR Door Speaker Is Inoperative

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell 130 , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to ACM in Information and Entertainment System.

Root DTC Description	Failure Types	Fault Trigger Conditions
B1A03:01	Speaker #3: General Electrical Failure	Set by the ACM when a failure is detected on the RR door speaker circuits.
B1A03:11	Speaker #3: Circuit Short to Ground	Set by the ACM when a short to ground is detected on the RR door speaker circuits. When DTC B1A03:11 is set, the speaker output is disabled.
B1A03:12	Speaker #3: Circuit Short to Battery	Set by the ACM when a short to battery is detected on the RR door speaker circuits. When DTC B1A03:12 is set, the speaker output is disabled.
B1A03:13	Speaker #3: Circuit Open	Set by the ACM when an open is detected on the RR door speaker circuits.

Possible Sources

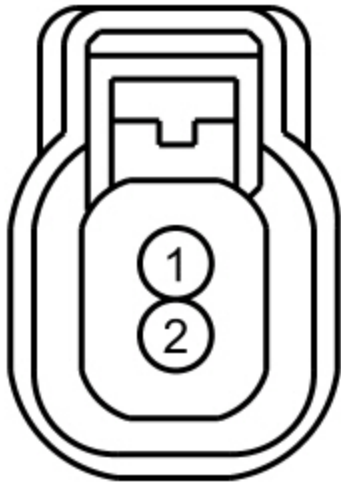
- Wiring, terminals, or connectors
- Speaker
- ACM

PINPOINT TEST J : THE RR DOOR SPEAKER IS INOPERATIVE

J1 CHECK THE AUDIO SIGNAL TO THE RR DOOR SPEAKER

- Ignition OFF.
- Disconnect: RR Door Speaker [C802](#).
- Operate the audio system in radio tuner AM/FM mode.
- Measure the **AC voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C802 Pin 1	VME12 (BN/WH)	C802 Pin 2	RME12 (BN/BU)



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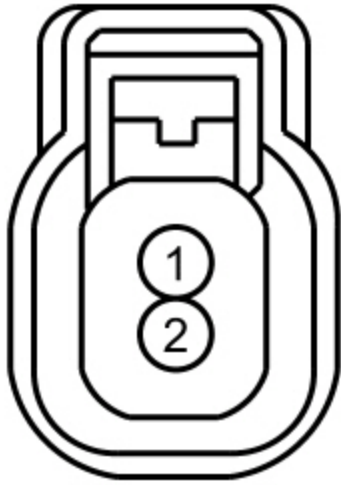
Is a fluctuating AC voltage present?

Yes	INSTALL a new RR door speaker. REFER to Speaker — Door.
No	GO to J2 .

J2 CHECK THE AUDIO CIRCUITS TO THE RR DOOR SPEAKER FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: ACM [C240A](#) .
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C802 Pin 1	VME12 (BN/WH)	—	Ground
C802 Pin 2	RME12 (BN/BU)	—	Ground



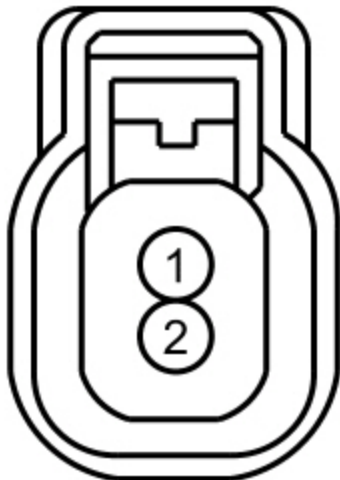
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to J3 .

J3 CHECK THE AUDIO CIRCUITS TO THE RR DOOR SPEAKER FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C802 Pin 1	VME12 (BN/WH)	—	Ground
C802 Pin 2	RME12 (BN/BU)	—	Ground



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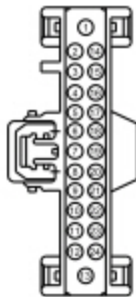
Are the resistances greater than 10,000 ohms?

Yes	GO to J4 .
No	REPAIR the circuit in question.

J4 CHECK THE AUDIO CIRCUITS TO THE RR DOOR SPEAKER FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C802 Pin 1	VME12 (BN/WH)	C240A Pin 24	VME12 (BN/WH)
C802 Pin 2	RME12 (BN/BU)	C240A Pin 12	RME12 (BN/BU)



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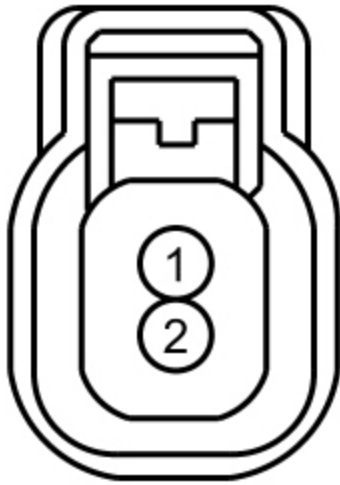
Are the resistances less than 3 ohms?

Yes	GO to J5 .
No	REPAIR the circuit in question.

J5 CHECK THE AUDIO CIRCUITS TO THE RR DOOR SPEAKER FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C802 Pin 1	VME12 (BN/WH)	C802 Pin 2	RME12 (BN/BU)



Are the resistances greater than 10,000 ohms?

Yes	GO to J6 .
No	REPAIR the circuit in question.

J6 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the ~~ACM~~ connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the ~~ACM~~ connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test K: The LR Door Speaker Is Inoperative

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to [ACM](#) in Information and Entertainment System.

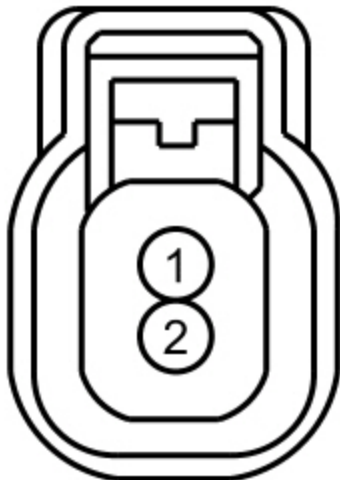
Root DTC Description	Failure Types	Fault Trigger Conditions
B1A04:01	Speaker #4: General Electrical Failure	Set by the ACM when a failure is detected on the LR door speaker circuits.
B1A04:11	Speaker #4: Circuit Short to Ground	Set by the ACM when a short to ground is detected on the LR door speaker circuits. When DTC B1A04:11 is set, the speaker output is disabled.
B1A04:12	Speaker #4: Circuit Short to Battery	Set by the ACM when a short to battery is detected on the LR door speaker circuits. When DTC B1A04:12 is set, the speaker output is disabled.
B1A04:13	Speaker #4: Circuit Open	Set by the ACM when an open is detected on the LR door speaker circuits.

Possible Sources

- Wiring, terminals, or connectors
- Speaker
- [ACM](#)

PINPOINT TEST K : THE LR DOOR SPEAKER IS INOPERATIVE

K1 CHECK THE AUDIO SIGNAL TO THE LR DOOR SPEAKER			
• Ignition OFF. • Disconnect: LR Door Speaker C702. • Operate the audio system in radio tuner AM/FM mode. • Measure the AC voltage between:			
Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C702 Pin 1	VME09 (WH/GN)	C702 Pin 2	RME09 (BN/YE)



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Is a fluctuating AC voltage present?

Yes	INSTALL a new LR door speaker. REFER to Speaker — Door.
No	GO to K2 .

K2 CHECK THE AUDIO CIRCUITS TO THE LR DOOR SPEAKER FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: ACM [C240A](#) .
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C702 Pin 1	VME09 (WH/GN)	—	Ground
C702 Pin 2	RME09 (BN/YE)	—	Ground



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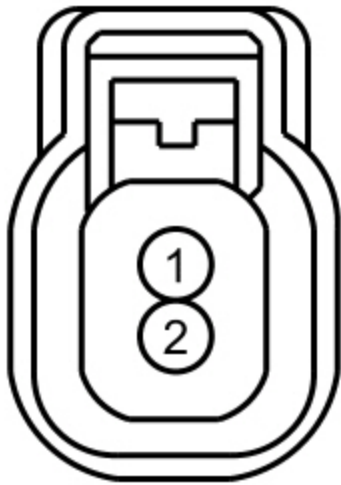
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to K3 .

K3 CHECK THE AUDIO CIRCUITS TO THE LR DOOR SPEAKER FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C702 Pin 1	VME09 (WH/GN)	—	Ground
C702 Pin 2	RME09 (BN/YE)	—	Ground



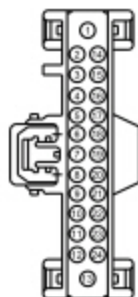
Are the resistances greater than 10,000 ohms?

Yes	GO to K4 .
No	REPAIR the circuit in question.

K4 CHECK THE AUDIO CIRCUITS TO THE LR DOOR SPEAKER FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C702 Pin 1	VME09 (WH/GN)	C240A Pin 21	VME09 (WH/GN)
C702 Pin 2	RME09 (BN/YE)	C240A Pin 9	RME09 (BN/YE)



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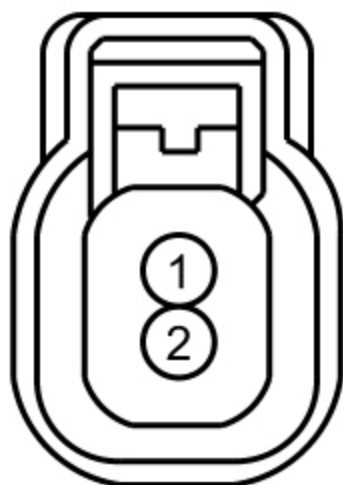
Are the resistances less than 3 ohms?

Yes	GO to K5 .
No	REPAIR the circuit in question.

K5 CHECK THE AUDIO CIRCUITS TO THE LR DOOR SPEAKER FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C702 Pin 1	VME09 (WH/GN)	C702 Pin 2	RME09 (BN/YE)



Are the resistances greater than 10,000 ohms?

Yes	GO to K6 .
No	REPAIR the circuit in question.

K6 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the **ACM** connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **ACM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test L: Poor Quality, Distorted, Or No Sound While In SYNC® Mode

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to SYNC® System in Information and Entertainment System.

The **APIM** transmits left and right channel analog audio signals to the **ACM** . These signals include inputs from the **USB** port, audio input jack, and Bluetooth media device. If a **USB** mass storage device is used to play an audio file, the SYNC® system only plays audio files that do not have **DRM** protection.

Possible Sources

- Customer's device
- Wiring, terminals, or connectors
- **APIM**
- **ACM**

PINPOINT TEST L : POOR QUALITY, DISTORTED, OR NO SOUND WHILE IN SYNC® MODE	
L1 VERIFY THE OPERATION OF THE SYNC® SYSTEM AUDIO SOURCES	
• Connect: Multi-Media Interface Tester 105-00120. • Using the Multi-Media Interface Tester 105-00120, test the audio output for the audio input jack, USB port, and Bluetooth.	

Is there poor quality, distorted, or no sound from all SYNC® audio sources?

Yes	GO to L2 .
No	If there is a concern with one or more, but not all audio sources, GO to Symptom Chart — SYNC® System to diagnose the observed symptom. If all audio sources operate correctly, the concern is with the customer's device.

L2 CHECK THE AUDIO SIGNALS TO THE ACM

NOTICE: This pinpoint test step directs testing circuits using a back-probe method. Use the special back-probe tool POM6411. Do not force test leads or other probes into connectors. Adequate care must be exercised to avoid connector terminal damage, while ensuring that good electrical contact is made with the circuit or terminal. Failure to follow these instructions may cause damage to wiring, terminals, or connectors, and may cause subsequent electrical faults.

- Operate the audio system in SYNC® mode and play an audio file.
- Measure the AC voltage by back-probing ACM connector [C240B](#):

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C240B Pin 12	VME52 (BU)	C240B Pin 25	RME52 (GY/OG)
C240B Pin 13	VME53 (VT/GN)	C240B Pin 26	RME53 (BN/WH)

Is a fluctuating AC voltage present?

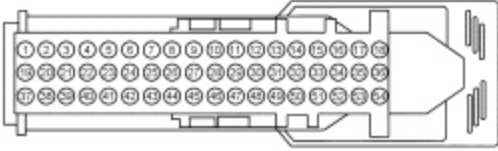
Yes	GO to L7 .
No	GO to L3 .

L3 CHECK THE AUDIO CIRCUITS TO THE ACM FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: ACM [C240B](#) .
- Disconnect: APIM [C2383](#) .
- Ignition ON.
- Measure the voltage between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2383 Pin 23	VME53 (VT/GN)	—	Ground
C2383 Pin 24	RME53 (BN/WH)	—	Ground

C2383 Pin 25	VME52 (BU)	—	Ground
C2383 Pin 26	RME52 (GY/OG)	—	Ground



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Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to L4 .

L4 CHECK THE AUDIO CIRCUITS TO THE ACM FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2383 Pin 23	VME53 (VT/GN)	—	Ground
C2383 Pin 24	RME53 (BN/WH)	—	Ground
C2383 Pin 25	VME52 (BU)	—	Ground
C2383 Pin 26	RME52 (GY/OG)	—	Ground



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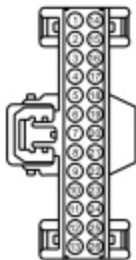
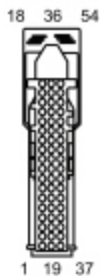
Are the resistances greater than 10,000 ohms?

Yes	GO to L5 .
No	REPAIR the circuit in question.

L5 CHECK THE CIRCUITS TO THE ACM FOR AN OPEN

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2383 Pin 23	VME53 (VT/GN)	C240B Pin 13	VME53 (VT/GN)
C2383 Pin 24	RME53 (BN/WH)	C240B Pin 26	RME53 (BN/WH)
C2383 Pin 25	VME52 (BU)	C240B Pin 12	VME52 (BU)
C2383 Pin 26	RME52 (GY/OG)	C240B Pin 25	RME52 (GY/OG)



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Are the resistances less than 3 ohms?

Yes	GO to L6 .
No	REPAIR the circuit in question.

L6 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.
- Disconnect and inspect the **APIM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **APIM** connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the **APIM** to re-initialize.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

L7 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the **ACM** connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **ACM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Normal Operation and Fault Conditions

Refer to SYNC® System in Information and Entertainment System.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
U3000:04	Control Module: System Internal Failure	Set by the APIM when it detects a fault due to a device conflict or internal failure.

Possible Sources

- Network communication concern
- Customer error
- Customer's device
- APIM

PINPOINT TEST M : ALL SYNC® FUNCTIONS ARE INOPERATIVE OR DO NOT OPERATE CORRECTLY

M1 VERIFY THAT THE APIM PASSES THE NETWORK TEST	
• Using a diagnostic scan tool, perform the network test.	
Did the APIM pass the network test?	
Yes	GO to M2 .
No	REFER to Section 418-00.
M2 CHECK THE SYNC® SYSTEM INPUTS FOR FUNCTIONALITY	
NOTE: <i>Carrying out a Master Reset returns all preference settings to the factory defaults, erases the phone book and call histories, and deletes any devices paired with the SYNC® system.</i>	
• Connect: Multi-Media Interface Tester 105-00120. • Using the Multi-Media Interface Tester 105-00120, test the audio output for the audio input jack, USB port, and Bluetooth connection using the VOICE switch on the steering wheel controls to enter each mode.	
Do all of the SYNC® inputs function correctly?	
Yes	The SYNC® system is operating correctly at this time. CARRY OUT a SYNC® system Master Reset. REFER to the Owner's Literature. REVIEW the SYNC® system operation with the customer. If the customer's device still does not operate correctly, the fault is with the customer's device.

No	GO to M3 .
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M3 RESET THE APIM AND RECHECK SYNC® SYSTEM OPERATION

NOTE: *Carrying out a Master Reset returns all preference settings to the factory defaults, erases the phone book and call histories, and deletes any devices paired with the SYNC® system.*

- Carry out an ~~APIM~~ power reset by disconnecting the battery for 1 minute, then reconnecting it.
- Connect: Multi-Media Interface Tester 105-00120.
- Using the Multi-Media Interface Tester 105-00120, test the audio output for the audio input jack, ~~USB~~ port, and Bluetooth connection using the VOICE switch on the steering wheel controls to enter each mode.

Do all of the SYNC® inputs function correctly?

Yes	The SYNC® system is operating correctly at this time. CARRY OUT a SYNC® system Master Reset. REFER to the Owner's Literature. REVIEW the SYNC® system operation with the customer. If the customer's device still does not operate correctly, the fault is with the customer's device.
No	If only some, but not all of the inputs are inoperative, GO to Symptom Chart — SYNC® System to diagnose the observed symptom. If all inputs are inoperative and/or DTC U3000:04 is present, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.

Pinpoint Test N: The SYNC® System Voice Or Tone Prompts, TTS Feature, Or Ringtones Are Inoperative Or Do Not Operate Correctly, Or During A Phone Call, No Incoming Audio Is Heard In The Vehicle

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System for schematic and connector information.

Normal Operation and Fault Conditions

Refer to SYNC® System and Voice Recognition in Information and Entertainment System.

Possible Sources

- Customer's setting
- Wiring, terminals, or connectors
- ~~APIM~~
- ~~ACM~~

PINPOINT TEST N : THE SYNC® SYSTEM VOICE OR TONE PROMPTS, TTS FEATURE, OR RINGTONES ARE INOPERATIVE OR DO NOT OPERATE CORRECTLY, OR DURING A PHONE CALL, NO INCOMING AUDIO IS HEARD IN THE VEHICLE

N1 CHECK THE AUDIBLE PROMPT SETTING

- Operate the audio system in SYNC® mode.

- Verify the audible prompts are enabled. Refer to "SYNC® Voice Recognition Feature" in the Owner's Literature.
- Operate the VOICE button on the steering wheel controls and observe the SYNC® audible prompt.

Does the SYNC® system produce an audible prompt correctly?

Yes	The system is operating correctly at this time. The concern was caused by the customer's setting. INSTRUCT the customer in the correct operation of the audible prompt feature.
No	GO to N2 .

N2 CHECK FOR A VOLTAGE SIGNAL FROM THE APIM

NOTICE: This pinpoint test step directs testing circuits using a back-probe method. Use the special back-probe tool POM6411. Do not force test leads or other probes into connectors. Adequate care must be exercised to avoid connector terminal damage while ensuring that good electrical contact is made with the circuit or terminal. Failure to follow these instructions may cause damage to wiring, terminals, or connectors, and may cause subsequent electrical faults.

- Operate the audio system in SYNC® mode.
- While operating the VOICE button repeatedly, measure the **AC voltage** by back-probing **ACM** connector [C240B](#):

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C240B Pin 8	VMN14 (WH/VT)	C240B Pin 21	RMN14 (GY/BN)

Is a fluctuating AC voltage present each time the VOICE switch is operated?

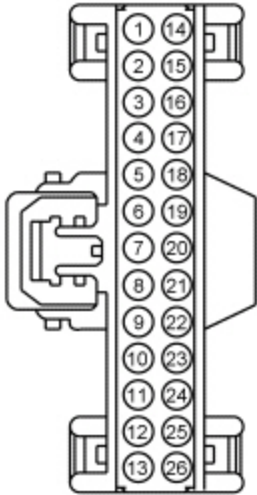
Yes	GO to N7 .
No	GO to N3 .

N3 CHECK THE AUDIBLE PROMPT CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: **ACM** [C240B](#) .
- Disconnect: **APIM** [C2383](#) .
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit

C240B Pin 8	VMN14 (WH/VT)	—	Ground
C240B Pin 21	RMN14 (GY/BN)	—	Ground



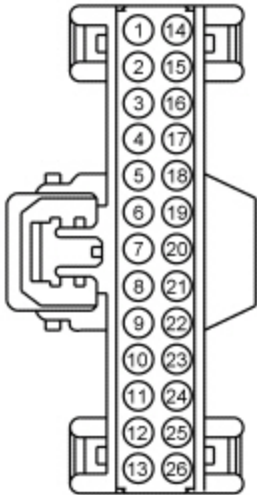
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to N4 .

N4 CHECK THE AUDIBLE PROMPT CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C240B Pin 8	VMN14 (WH/VT)	—	Ground
C240B Pin 21	RMN14 (GY/BN)	—	Ground



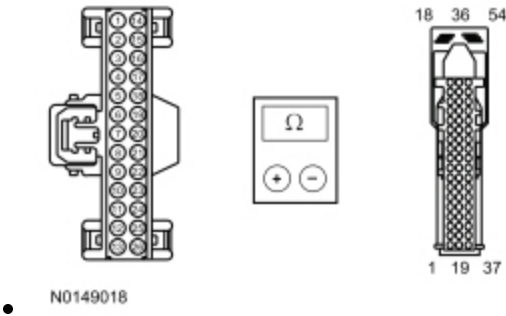
Are the resistances greater than 10,000 ohms?

Yes	GO to N5 .
No	REPAIR the circuit in question.

N5 CHECK THE AUDIBLE PROMPT CIRCUITS FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C240B Pin 8	VMN14 (WH/VT)	C2383 Pin 3	VMN14 (WH/VT)
C240B Pin 21	RMN14 (GY/BN)	C2383 Pin 4	RMN14 (GY/BN)



Are the resistances less than 3 ohms?

Yes	GO to N6 .
No	REPAIR the circuit in question.

N6 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.
- Disconnect and inspect the APIM connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the APIM connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the APIM to re-initialize.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

N7 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the ~~ACM~~ connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the ~~ACM~~ connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test O: All SYNC® Services Features Are Inoperative Or Inaccurate

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Normal Operation and Fault Conditions

Refer to SYNC® Services in Information and Entertainment System.

Possible Sources

- Expired subscription
- ~~APIM~~
- ~~GPSM~~
- SYNC® Services server

PINPOINT TEST O : ALL SYNC® SERVICES FEATURES ARE INOPERATIVE OR INACCURATE

O1 VERIFY THE CUSTOMER'S SYNC® OWNER ACCOUNT AND SYNC® SERVICES REGISTRATION AND SUBSCRIPTION

- Verify that the customer has a SYNC® Owner Account on [SyncMyRide](#)
- Verify that the customer has registered for SYNC® Services and has an active subscription.

Does the customer have a SYNC® Owner Account and an active SYNC® Services subscription?

Yes	GO to O2 .
No	If the customer does not have a SYNC® Owner Account, INFORM the customer that an account must be created in order to sign up for SYNC® Services. REFER to SyncMyRide . If the customer does not have an active SYNC® Services subscription, INFORM the customer that in order for SYNC® Services to function, the subscription needs to be activated. REFER to SyncMyRide .

O2 VERIFY THE CUSTOMER'S REGISTERED PHONE NUMBER AND STATUS

- Verify that the mobile phone being used has the correct phone number registered with SYNC® Services.
- Verify that the mobile phone being used is selected as "Active" on [SyncMyRide](#).

Is the customer using the registered and active mobile phone associated with the SYNC® services account?

Yes	GO to O3 .
No	INFORM the customer that the correct phone number must be registered on the SYNC® website and selected as "Active." REFER to SyncMyRide .

O3 CHECK FOR APIM DTCS

- Ignition ON.
- Using a diagnostic scan tool, perform the **APIM** self-test.

Are any DTCs recorded in the **APIM ?**

Yes	REFER to the APIM DTC Chart in Information and Entertainment System.
No	GO to O4 .

O4 CHECK FOR GPSM DTCS

- Using a diagnostic scan tool, perform the **GPSM** self-test.

Are any DTCs recorded in the **GPSM ?**

Yes	REFER to the GPSM DTC Chart in Information and Entertainment System.
No	GO to O5 .

O5 VERIFY THE CUSTOMER'S PHONE CONNECTION TO SYNC® SERVICES

- Advise the customer to connect their phone to SYNC® Services. Caller ID blocking must be disabled on the customer's phone. Refer to the Owner's Literature.
- Operate a SYNC® Services feature.
- Verify that the phone call is placed and that the phone is connected to SYNC® Services.

Does the phone connect to SYNC® Services and operate correctly?

Yes	GO to O6 .
No	GO to Pinpoint Test S .

O6 OBSERVE THE SYNC® SERVICES SUBSCRIPTION COMMERCIAL

- Once the customer's phone is connected to SYNC® Services, observe the subscription commercial.

Does the SYNC® Services subscription commercial repeat with no selection option?

Yes	There is a potential SYNC® Services server issue, and the problem does not reside in the vehicle's audio system. CONTACT the Ford Technical Hotline for assistance.
No	GO to O7 .

O7 CHECK THE OPERATION OF THE DIRECTIONS FROM SYNC® SERVICES

- When prompted, select directions from SYNC® Services and observe the audible prompts from the audio system.

Is a single chime, audible prompt, then a short chime heard, indicating the navigation route is being sent to the vehicle?

Yes	SYNC® Services is operating correctly at this time.
No	If a single chime, audible prompt, then a repeating chime is heard, indicating the navigation route is being sent to the vehicle, GO to O8 . If no chime or audible prompt is heard, indicating that the system was unable to locate the vehicle, there is a potential SYNC® Services server issue. The problem does not reside in the vehicle's audio system. CONTACT the Ford Technical Hotline for assistance.

O8 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.
- Disconnect and inspect the APIM connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)

- Pushed-out pins (install new pins as necessary)

- Reconnect the **APIM** connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the **APIM** to re-initialize.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test P: Voice Recognition Is Inoperative Or Does Not Operate Correctly, Or During A Phone Call, Poor Quality Or No Outgoing Audio Is Heard On The Outside Device

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System for schematic and connector information.

Normal Operation and Fault Conditions

Refer to Voice Recognition in Information and Entertainment System.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
B1D79:01	Microphone Input: General Electrical Failure	Set by the APIM when a microphone is not detected during the on-demand self-test.
B1D79:11	Microphone Input: Circuit Short To Ground	Set by the APIM when short to ground is detected on the microphone circuits.
B1D79:12	Microphone Input: Circuit Short To Battery	Set by the APIM when short to voltage is detected on the microphone circuits.
B1D79:13	Microphone Input: Circuit Open	Set by the APIM when an open is detected on the microphone circuits.

Possible Sources

- Insufficiently quiet vehicle
- Inadequate phone reception

- Wiring, terminals, or connectors
- Microphone
- APIM

PINPOINT TEST P : VOICE RECOGNITION IS INOPERATIVE OR DOES NOT OPERATE CORRECTLY, OR DURING A PHONE CALL, POOR QUALITY OR NO OUTGOING AUDIO IS HEARD ON THE OUTSIDE DEVICE

P1 CHECK THE AUDIBLE PROMPT FUNCTIONALITY

- Operate the audio system in radio tuner AM/FM mode.
- Operate the VOICE steering wheel switch and listen for the SYNC® audible prompt.

Does the SYNC® system produce an audible prompt correctly, indicating voice recognition mode has been entered?

Yes	GO to P3 .
No	GO to P2 .

P2 CHECK THE OPERATION OF THE RH STEERING WHEEL SWITCHES

- Operate the audio system in AM/FM mode.
- Operate the VOLUME +/-, SEEK +/-, MEDIA, PHONE, and OK steering wheel switches.

Do the VOLUME +/-, SEEK +/-, MEDIA, PHONE, and OK steering wheel switches operate correctly?

Yes	GO to Pinpoint Test N .
No	GO to Pinpoint Test T to diagnose the RH steering wheel switches.

P3 CHECK THE OPERATION OF THE "PHONE" VOICE COMMAND

- Pair a Bluetooth phone with the SYNC® system. For information on the pairing process, refer to the Owner's Literature. If necessary, review the device compatibility list found on syncmyride.com.
- Operate the VOICE steering wheel switch and listen for the SYNC® audible prompt.
- **NOTE:** *Make sure the interior of the vehicle is as quiet as possible. Excessive noise may prevent the system from correctly recognizing spoken commands.*

Clearly speak the word "phone."

Does the SYNC® system respond correctly to the "phone" voice command?

Yes	For a concern with the voice recognition being inoperative or not operating correctly, the system is operating correctly at this time. The concern may have been caused by an insufficiently quiet vehicle. Delete the just-paired Bluetooth phone from the SYNC® system. For information on deleting a Bluetooth device from the SYNC® system, refer to the Owner's Literature. For a concern with poor quality or no outgoing audio being heard on the outside device during a phone call, GO to P4
No	GO to P5 .

P4 CHECK THE OUTGOING PHONE CALL QUALITY AND VOICE VOLUME

- Ensure that the Bluetooth phone paired in step 3 has adequate reception.
- **NOTE:** *Make sure the interior of the vehicle is as quiet as possible. Excessive noise may prevent the call from having clear call quality and acceptable voice volume.*

Using the SYNC® system, place a phone call. For information on making calls from the SYNC® system, refer to the Owner's Literature.

- Ensure that the called party's phone has adequate reception.
- Instruct the called party to determine if the call quality is clear and of acceptable volume.
- Delete the Bluetooth phone paired in step 3 from the SYNC® system. For information on deleting a Bluetooth device from the SYNC® system, refer to the Owner's Literature.

Is the outgoing call quality clear and of acceptable volume?

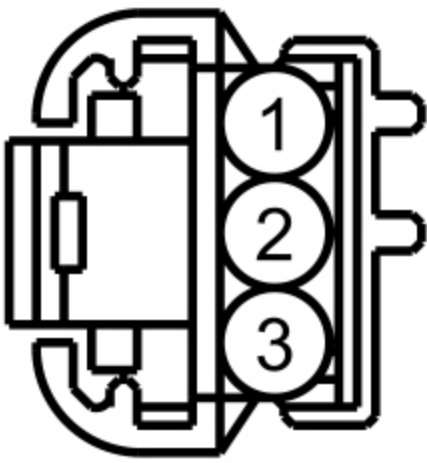
Yes	The system is operating correctly at this time. The concern may have been caused by poor phone reception or an insufficiently quiet vehicle.
No	GO to P5 .

P5 CHECK THE MICROPHONE AND MICROPHONE CIRCUITS FOR CORRECT VOLTAGE DROP

NOTICE: This pinpoint test step directs testing circuits using a back-probe method. Use the special back-probe tool POM6411. Do not force test leads or other probes into connectors. Adequate care must be exercised to avoid connector terminal damage, while ensuring that good electrical contact is made with the circuit or terminal. Failure to follow these instructions may cause damage to wiring, terminals, or connectors, and may cause subsequent electrical faults.

- Ignition ON.
- While back-probing microphone [C9221](#), measure and record the **DC voltage**.

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C9221 Pin 2	VMM13 (YE/GN)	C9221 Pin 3	DMM13 (BK)



Is the voltage between 2.7 and 6.2 volts?

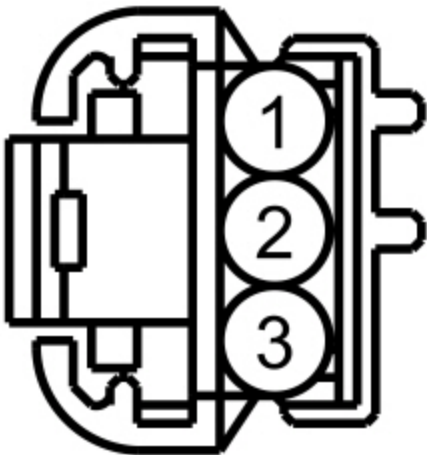
Yes	If diagnosing an intermittent concern, GO to P6 . If not diagnosing an intermittent concern, GO to P12 .
No	GO to P7 .

P6 CHECK THE MICROPHONE CIRCUITS FOR INTERMITTENT CONCERNS

NOTICE: This pinpoint test step directs testing circuits using a back-probe method. Use the special back-probe tool POM6411. Do not force test leads or other probes into connectors. Adequate care must be exercised to avoid connector terminal damage, while ensuring that good electrical contact is made with the circuit or terminal. Failure to follow these instructions may cause damage to wiring, terminals, or connectors, and may cause subsequent electrical faults.

- While back-probing microphone [C9221](#), measure the **DC voltage** while agitating the microphone harness back and forth and up and down (wiggle testing).

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C9221 Pin 2	VMM13 (YE/GN)	C9221 Pin 3	DMM13 (BK)



- Review the recorded results from step 5.

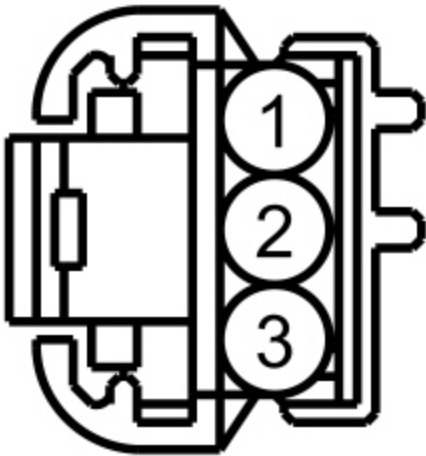
Does the voltage fluctuate more than .5 volts from the recorded result in step 5?

Yes	REPAIR the circuit(s), connector, or terminal(s) in question. For corrosion, clean pins or install a new connector or terminals. For damaged or bent pins, install new terminals or pins.
No	GO to P11 .

P7 CHECK THE MICROPHONE CIRCUITS TO THE APIM FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: Microphone [C9221](#).
- Disconnect: APIM [C2383](#).
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C9221 Pin 2	VMM13 (YE/GN)	—	Ground
C9221 Pin 3	DMM13 (BK)	—	Ground



Is any voltage present?

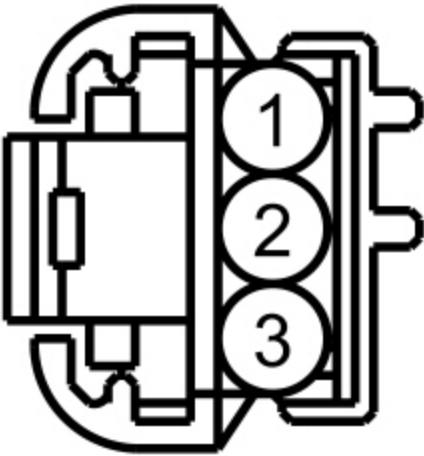
Yes	REPAIR the circuit in question.
No	GO to P8 .

P8 CHECK THE MICROPHONE CIRCUITS TO THE APIM FOR A SHORT TO GROUND

- Ignition OFF.

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C9221 Pin 2	VMM13 (YE/GN)	—	Ground
C9221 Pin 3	DMM13 (BK)	—	Ground



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Are the resistances greater than 10,000 ohms?

Yes	GO to P9 .
No	REPAIR the circuit in question.

P9 CHECK THE MICROPHONE CIRCUITS TO THE APIM FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C9221 Pin 2	VMM13 (YE/GN)	C2383 Pin 5	VMM13 (YE/GN)
C9221 Pin 3	DMM13 (BK)	C2383 Pin 6	DMM13 (BK)



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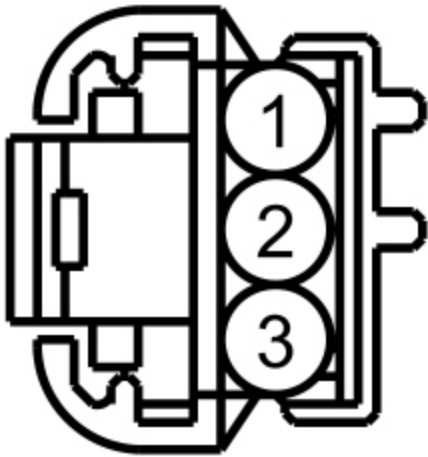
Are the resistances less than 3 ohms?

Yes	GO to P10 .
No	REPAIR the circuit in question.

P10 CHECK THE MICROPHONE CIRCUITS TO THE APIM FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C9221 Pin 2	VMM13 (YE/GN)	C9221 Pin 3	DMM13 (BK)



Is the resistance greater than 10,000 ohms?

Yes	GO to P12 .
No	REPAIR the circuit in question.

P11 ISOLATE THE MICROPHONE

- Install a new microphone. Refer to Microphone.
- If necessary, connect: APIM [C2383](#).
- Operate the audio system in AM/FM mode.
- Operate the VOICE steering wheel switch and listen for the SYNC® audible prompt.
- Speak a command to test the voice recognition for normal operation.

Is the voice recognition functioning correctly?

Yes	The system is operating correctly at this time. The concern was caused by an inoperative microphone.
No	GO to P12 .

P12 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.
- Disconnect and inspect the **APIM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **APIM** connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the **APIM** to re-initialize.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test Q: The Audio Input Jack Is Inoperative Or Has Poor Sound Quality

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to Audio Input Jack Mode in Information and Entertainment System.

Possible Sources

- Customer's device
- Wiring, terminals, or connectors
- Audio input jack
- **APIM**

PINPOINT TEST Q : THE AUDIO INPUT JACK IS INOPERATIVE OR HAS POOR SOUND QUALITY	
NOTE: Before carrying out this pinpoint test, make sure the MP3 device is operating correctly.	

Q1 CHECK THE AUDIO INPUT JACK AUDIO FUNCTIONALITY

- Connect: Multi-Media Interface Tester 105-00120.
- Using the Multi-Media Interface Tester 105-00120, attempt to play an audio file using the audio input jack.

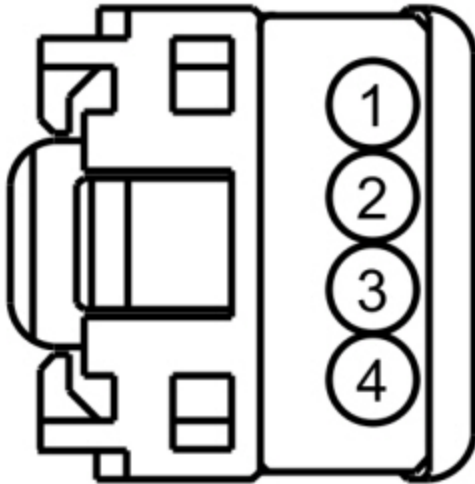
Is the audio output OK for the audio input jack?

Yes	The system is operating correctly at this time. The concern may be with the customer's device.
No	GO to Q2 .

Q2 CHECK THE AUDIO INPUT JACK CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: Audio Input Jack [C2362](#).
- Disconnect: ~~APIM~~ [C2383](#).
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2362 Pin 1	VME46 (BU/GN)	—	Ground
C2362 Pin 3	RME45 (YE/GN)	—	Ground
C2362 Pin 4	VME45 (BU)	—	Ground



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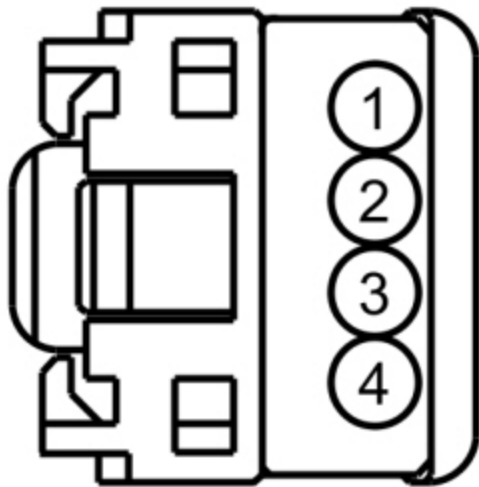
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to Q3 .

Q3 CHECK THE AUDIO INPUT JACK CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2362 Pin 1	VME46 (BU/GN)	—	Ground
C2362 Pin 3	RME45 (YE/GN)	—	Ground
C2362 Pin 4	VME45 (BU)	—	Ground



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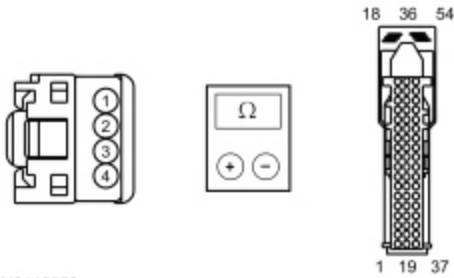
Is the resistance greater than 10,000 ohms?

Yes	GO to Q4 .
No	REPAIR the circuit.

Q4 CHECK THE AUDIO INPUT JACK CIRCUITS FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2362 Pin 1	VME46 (BU/GN)	C2383 Pin 47	VME46 (BU/GN)
C2362 Pin 3	RME45 (YE/GN)	C2383 Pin 46	RME45 (YE/GN)
C2362 Pin 4	VME45 (BU)	C2383 Pin 45	VME45 (BU)



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Are the resistances less than 3 ohms?

Yes	GO to Q5 .
No	REPAIR the circuit in question.

Q5 ISOLATE THE AUDIO INPUT JACK

- Connect: APIM [C2383](#).
- Install a new audio input jack. Refer to Audio Input Jack.
- Connect: Multi-Media Interface Tester 105-00120.
- Using the Multi-Media Interface Tester 105-00120, attempt to play an audio file using the audio input jack.

Is the audio output OK for the audio input jack?

Yes	The system is now correctly at this time. The concern was caused by an inoperative audio input jack.
No	GO to Q6 .

Q6 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.
- Disconnect and inspect the APIM connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the APIM connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the APIM to re-initialize.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test R: The USB Port Is Inoperative Or Does Not Operate Correctly

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to SYNC® AppLink™ , USB Audio Mode and USB Port. in Information and Entertainment System.

The USB port is part of the USB cable, and cannot be replaced separately.

The Multi-Media Interface Tester 105-00120 is powered by the USB port

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
B1252:04	USB Port: System Internal Failure	Set by the APIM when an over-temperature condition is detected in the USB circuit. This can be caused by a fault in the USB cable, USB port, or customer's USB device.
B1252:11	USB Port: Circuit Short to Ground	Set by the APIM when a short to ground or over-current condition is detected in the USB circuit. This can be caused by a fault in the USB cable, USB port, or customer's

DTC	Description	Fault Trigger Conditions
		USB device.

Possible Sources

- Customer's device
- USB cable/port
- USB cable/port connection
- APIM

PINPOINT TEST R : THE USB PORT IS INOPERATIVE OR DOES NOT OPERATE CORRECTLY

R1 RESET THE APIM AND RECHECK SYNC® SYSTEM OPERATION

- Remove any devices connected to the USB port.
- Carry out an APIM power reset by disconnecting the battery for 1 minute, then reconnecting it.
- Wait at least 2 minutes for the APIM to re-initialize.
- Connect: Multi-Media Interface Tester 105-00120.
- Using the Multi-Media Interface Tester 105-00120, attempt to play an audio file using the USB port.

Is the audio output OK for the USB port?

Yes	The system is operating correctly at this time. The concern was corrected by performing the APIM power reset.
No	GO to R2.

R2 CHECK THE APIM FOR DTC B1252:04 OR DTC B1252:11

- Using a diagnostic scan tool, clear the APIM DTCs.
- Perform the APIM self-test.

Is DTC B1252:04 or DTC B1252:11 present?

Yes	GO to R3.
No	GO to R5.

R3 INSPECT THE USB CABLE AND APIM

- Ignition OFF.
- Disconnect: USB cable at the APIM .
- Inspect the APIM connector port and both ends of the USB cable for damage. Remove any debris.

Is any damage present?

Yes	For the USB cable, INSTALL a new USB cable. REFER to Universal Serial Bus (USB) Cable and Port. For the APIM , INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	GO to R4 .

R4 CHECK THE USB PORT FOR FUNCTIONALITY

- Connect: **USB** cable at the **APIM** .
- Connect: Multi-Media Interface Tester 105-00120.
- Using the Multi-Media Interface Tester 105-00120, attempt to play an audio file using the **USB** port.

Is the audio output OK for the **USB port?**

Yes	The system is operating correctly at this time. The concern was caused by the USB connector not being seated correctly.
No	GO to R5 .

R5 ISOLATE THE USB CABLE/PORT

- Install a new **USB** cable/port. Universal Serial Bus (USB) Cable and Port.
- Connect: Multi-Media Interface Tester 105-00120.
- Using the Multi-Media Interface Tester 105-00120, attempt to play a file using the **USB** port.

Is the audio output OK for the **USB port?**

Yes	The system is operating correctly at this time. The concern was caused by an inoperative USB cable.
No	GO to R6 .

R6 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.
- Disconnect and inspect the **APIM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **APIM** connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the **APIM** to re-initialize.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test S: Unable To Pair Bluetooth Device

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Normal Operation and Fault Conditions

Refer to Bluetooth Mode and SYNC® AppLink™ in Information and Entertainment System.

Possible Sources

- Incompatible Bluetooth device
- ~~APIM~~

PINPOINT TEST S : UNABLE TO PAIR BLUETOOTH DEVICE

S1 CHECK THE BLUETOOTH CONNECTION

NOTE: Carrying out a Master Reset returns all preference settings to the factory defaults, erases the phone book and call histories, and deletes any devices paired with the SYNC® system.

- Connect: Multi-Media Interface Tester 105-00120.
- Using the Multi-Media Interface Tester 105-00120, connect to the SYNC® system using Bluetooth. Follow the tool instructions.
- Using a diagnostic scan tool, view the ~~APIM~~ PIDs.
- Monitor the Bluetooth device paired PID (BT_PAIR) and the Bluetooth device connected PID (BT_CONN).

Do the PIDs both read "Yes"?

Yes	The SYNC® system is operating correctly at this time. CARRY OUT a SYNC® system Master Reset. REFER to the Owner's Literature. REVIEW the pairing process with the customer. If the customer's device still does not pair, the fault is with the customer's device.
No	GO to S2 .

S2 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.
- Disconnect and inspect the ~~APIM~~ connector.

- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **APIM** connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the **APIM** to re-initialize.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test T: The **RH steering wheel controls are inoperative or do not operate correctly**

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to Steering Wheel Switch Functions and Steering Wheel Switches in Information and Entertainment System.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
B1380:09	Steering Wheel Right Switch Pack: Component Failure	Set by the SCCM when the RH steering wheel switch is stuck.
B1380:11	Steering Wheel Right Switch Pack: Circuit Short To Ground	Set by the SCCM when a short to ground is detected in the RH steering wheel switch circuits.
B1380:17	Steering Wheel Right Switch Pack: Circuit Voltage Above Threshold	Set by the SCCM when a short to voltage is detected in the RH steering wheel switch circuits.

Possible Sources

- Steering wheel switch

- Wiring, terminals, or connectors
- Clockspring
- SCCM

PINPOINT TEST T : THE RH STEERING WHEEL CONTROLS ARE INOPERATIVE OR DO NOT OPERATE CORRECTLY

T1 CHECK THE RH STEERING WHEEL SWITCH PIDS

- Ignition ON.
- Monitor the steering wheel controls PIDs while pressing each **RH** steering wheel switch.
- Using a diagnostic scan tool, view the **SCCM** PIDs.
 - SW_VOLDOWN
 - SW_VOLUP
 - SW_SEEKMINUS
 - SW_SEEKPLUS
 - SW_VOICE
 - SW_MEDIA
 - SW_PHONE
 - SW_OK

Do the PID values agree with the switch positions?

Yes	GO to T10 .
No	If one or more, but not all PID values are incorrect or always read one particular switch position, INSTALL a new RH steering wheel switch. REFER to Section 211-05. If all PID values are incorrect or always read one particular switch position, GO to T2 .

T2 CHECK FOR IPC DTC U0212:00

- Ignition ON.
- Using a diagnostic scan tool, clear the DTCs from the **IPC**.
- Ignition OFF.
- Ignition ON.
- Wait at least 10 seconds.
- Using a diagnostic scan tool, perform the **IPC** self-test.

Is DTC U0212:00 set in the **IPC ?**

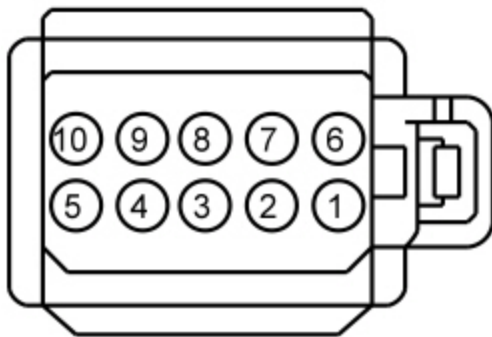
Yes	REFER to Section 413-01.
No	GO to T3 .

T3 CHECK THE CIRCUITRY BETWEEN THE CLOCKSPRING AND STEERING WHEEL CONTROLS FOR A SHORT TO GROUND

WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.

- Ignition OFF.
- Disconnect: RH Steering Wheel Switch [C2999](#).
- Disconnect: Clockspring [C218B](#).
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2999 Pin 2	VME14	—	Ground
C2999 Pin 4	RME14	—	Ground
C2999 Pin 6	VME54	—	Ground
C2999 Pin 7	RME54	—	Ground



-

Are the resistances greater than 10,000 ohms?

Yes	GO to T4 .
No	INSTALL a new steering wheel. REFER to Section 211-04.

T4 CHECK THE CIRCUITRY BETWEEN THE CLOCKSPRING AND STEERING WHEEL CONTROLS FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2999 Pin 2	VME14	C218B Pin 14	VME14
C2999 Pin 4	RME14	C218B Pin 6	RME14
C2999 Pin 6	VME54	C218B Pin 16	VME54
C2999 Pin 7	RME54	C218B Pin 8	RME54



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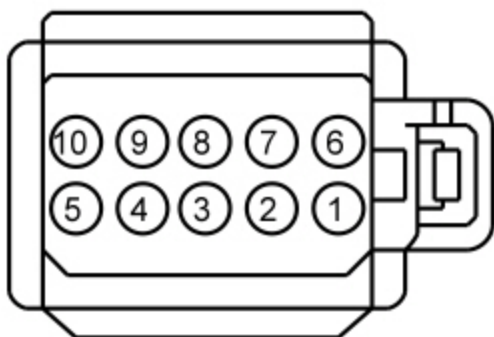
Are the resistances less than 3 ohms?

Yes	GO to T5 .
No	INSTALL a new steering wheel. REFER to Section 211-04.

T5 CHECK THE CIRCUITRY BETWEEN THE CLOCKSPRING AND STEERING WHEEL CONTROLS FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2999 Pin 2	VME14	C2999 Pin 4	RME14
C2999 Pin 6	VME54	C2999 Pin 7	RME54



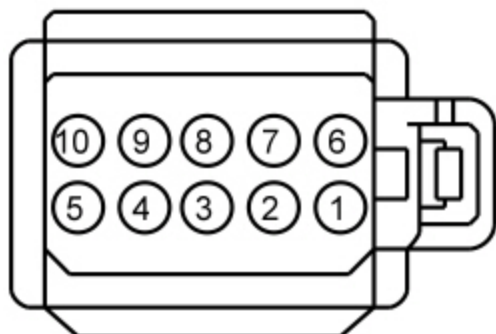
Are the resistances greater than 10,000 ohms?

Yes	GO to T6 .
No	INSTALL a new steering wheel. REFER to Section 211-04.

T6 CHECK THE CLOCKSPRING FOR A SHORT TO GROUND

- Connect: Clockspring [C218B](#).
- Disconnect: Clockspring [C218C](#).
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2999 Pin 2	VME14	—	Ground
C2999 Pin 4	RME14	—	Ground
C2999 Pin 6	VME54	—	Ground
C2999 Pin 7	RME54	—	Ground



-

Are the resistances greater than 10,000 ohms?

Yes	GO to T7 .
No	INSTALL a new clockspring. REFER to Section 501-20B.

T7 CHECK THE CLOCKSPRING FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2999 Pin 2	VME14	C218C Pin 6	—
C2999 Pin 4	RME14	C218C Pin 5	—
C2999 Pin 6	VME54	C218C Pin 10	—

C2999 Pin 7	RME54	C218C Pin 9	—
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- N0148946

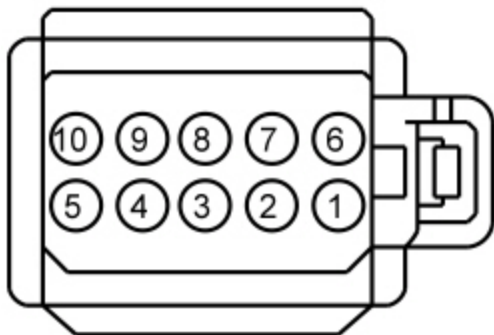
Are the resistances less than 3 ohms?

Yes	GO to T8 .
No	INSTALL a new clockspring. REFER to Section 501-20B.

T8 CHECK THE CLOCKSPRING FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2999 Pin 2	VME14	C2999 Pin 4	RME14
C2999 Pin 6	VME54	C2999 Pin 7	RME54



-

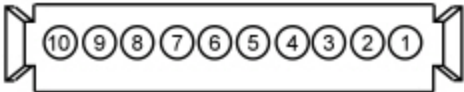
Are the resistances greater than 10,000 ohms?

Yes	GO to T9 .
No	INSTALL a new clockspring. REFER to Section 501-20B.

T9 CHECK FOR VOLTAGE AT THE SCCM

- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2414D Pin 6	—	C2414D Pin 5	—
C2414D Pin 10	—	C2414D Pin 9	—



-

Is approximately 5 volts present?

Yes	INSTALL a new RH steering wheel switch. REFER to Section 211-05.
No	GO to T10 .

T10 CHECK FOR CORRECT SCCM OPERATION

- Ignition OFF.
- Disconnect and inspect all the **SCCM** connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **SCCM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new SCCM . REFER to Section 211-05.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test U: The Audio System Does Not Operate Correctly From The FCIM

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Normal Operation and Fault Conditions

Refer to Accessory Delay Feature and ~~FCIM~~ in Information and Entertainment System.

Possible Sources

- Network communication concern
- ~~FCIM~~

PINPOINT TEST U : THE AUDIO SYSTEM DOES NOT OPERATE CORRECTLY FROM THE FCIM
U1 VERIFY THAT THE ACM , APIM , AND FCIM PASS THE NETWORK TEST

- Using a diagnostic scan tool, perform the network test.

Did the ~~ACM~~ , ~~APIM~~ , and ~~FCIM~~ pass the network test?

Yes	GO to U2 .
No	If the ACM failed the network test, REFER to Section 418-00 The ACM Does Not Respond To The Scan Tool. If the APIM failed the network test, REFER to Section 418-00 The APIM Does Not Respond To The Scan Tool. If the FCIM failed the network test, REFER to Section 418-00 The FCIM Does Not Respond To The Scan Tool.

U2 RESET THE AUDIO SYSTEM MODULES AND RECHECK THE FCIM OPERATION

- Disconnect the battery for 5 minutes, then reconnect it.
- Turn the infotainment system on.
- Test the ~~FCIM~~ for normal operation.

Does the ~~FCIM~~ operate correctly?

Yes	The FCIM is operating correctly at this time. The concern was corrected by resetting the audio system modules.
No	GO to U3

U3 RETRIEVE THE ACM , APIM , AND FCIM DTCS

- Using a diagnostic scan tool, perform the self-test for the following modules:
 - ACM
 - APIM
 - FCIM

Are any DTCs present in the ACM , APIM , or FCIM ?

Yes	For ACM DTCs, REFER to the ACM DTC Chart in Information and Entertainment System For APIM DTCs, REFER to the APIM DTC Chart in Information and Entertainment System For FCIM DTCs, REFER to the FCIM DTC Chart in Information and Entertainment System
No	GO to U4 .

U4 CHECK FOR CORRECT FCIM OPERATION

- Ignition OFF.
- Disconnect and inspect the FCIM connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the FCIM connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new FCIM . REFER to Front Controls Interface Module (FCIM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test V: The FCDIM Is Inoperative Or Does Not Operate Correctly

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Normal Operation and Fault Conditions

Refer to FCDIM in Information and Entertainment System.

Possible Sources

- Wiring, terminals, or connectors
- FCDIM

PINPOINT TEST V : THE FCDIM IS INOPERATIVE OR DOES NOT OPERATE CORRECTLY

V1 VERIFY THAT THE FCDIM PASSES THE NETWORK TEST

- Using a diagnostic scan tool, perform the network test.

Did the FCDIM pass the network test?

Yes	GO to V2 .
No	REFER to Section 418-00 The FCDIM Does Not Respond To The Scan Tool.

V2 OBSERVE THE FCDIM DISPLAY

- Turn the audio system on and verify the FCDIM operation.

Is the FCDIM screen completely inoperative or blank?

Yes	GO to V4 .
No	GO to V3 .

V3 CHECK FOR LOST COMMUNICATION DTCS SET IN THE FCDIM

- Ignition ON.
- Using a diagnostic scan tool, perform the FCDIM self-test.

Are any lost communication DTCS (U-codes) present in the FCDIM ?

Yes	REFER to the FCDIM DTC Chart in Information and Entertainment System.
No	GO to V4 .

V4 CHECK FOR CORRECT FCDIM OPERATION

- Ignition OFF.
- Disconnect and inspect the FCDIM connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the FCDIM connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new FCDIM . REFER to Front Controls/Display Interface Module (FCDIM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test W: One Or More FCDIM Displays Are Inoperative Or Do Not Operate Correctly

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Normal Operation and Fault Conditions

Refer to **FCDIM** in Information and Entertainment System.

All of the displays on the FCDIM screen are controlled through network messages. The modules sending display data to the FCDIM are:

- **ACM** : audio system information
- **APIM** : SYNC® system information
- **IPC** : warning messages and indicators

If the **FCDIM** loses the signal from any of these modules, only those specific displays will be inoperative.

Possible Sources

- **ACM**
- **APIM**
- **FCDIM**
- **IPC**

PINPOINT TEST W : ONE OR MORE FCDIM DISPLAYS ARE INOPERATIVE OR DO NOT OPERATE CORRECTLY

W1 VERIFY THAT THE FCDIM PASSES THE NETWORK TEST	
• Using a diagnostic scan tool, perform the network test.	
Did the FCDIM pass the network test?	
Yes	GO to W2 .
No	REFER to Section 418-00 The FCDIM Does Not Respond To The Scan Tool.
W2 CHECK FOR LOST COMMUNICATION DTCS SET IN THE FCDIM	
• Ignition ON. • Using a diagnostic scan tool, perform the FCDIM self-test.	
Are any lost communication DTCs (U-codes) present in the FCDIM ?	

Yes	REFER to the FCDIM DTC Chart in Information and Entertainment System.
No	GO to W3 .

W3 CHECK THE FCDIM DISPLAY SEGMENTS

- Using a diagnostic scan tool, view the FCDIM PIDs.
- Command all of the LCD segments on using the FCDIM active command LCD_SEG (All LCD Segments).

Do all of the FCDIM segments illuminate?

Yes	The system is operating correctly at this time. INFORM the customer of the operation of the FCDIM display.
No	GO to W4 .

W4 CHECK FOR CORRECT FCDIM OPERATION

- Ignition OFF.
- Disconnect and inspect the FCDIM connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the FCDIM connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new FCDIM . REFER to Front Controls/Display Interface Module (FCDIM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test X: The Compass Is Inoperative

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Normal Operation and Fault Conditions

Refer to FCDIM in Information and Entertainment System.

Possible Sources

- Network communication concern
- FCDIM
- GPSM

PINPOINT TEST X : THE COMPASS IS INOPERATIVE

X1 VERIFY THAT THE FCDIM AND GPSM PASS THE NETWORK TEST

- Using a diagnostic scan tool, perform the network test.

Did the FCDIM and GPSM pass the network test?

Yes	GO to X2 .
No	If the FCDIM failed the network test, REFER to Section 418-00 The FCDIM Does Not Respond To The Scan Tool. If the GPSM failed the network test, REFER to Section 418-00 The GPSM Does Not Respond To The Scan Tool.

X2 RETRIEVE THE FCDIM AND GPSM DTCS

- Using a diagnostic scan tool, perform the self-test for the following modules:
 - FCDIM
 - GPSM

Are any DTCs present in the FCDIM or GPSM ?

Yes	For FCDIM DTCs, REFER to the FCDIM DTC Chart in Information and Entertainment System For GPSM DTCs, REFER to the GPSM DTC Chart in Information and Entertainment System
No	GO to X3 .

X3 CHECK FOR CORRECT GPSM OPERATION

- Ignition OFF.
- Disconnect and inspect the GPSM connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the GPSM connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new GPSM . REFER to Global Positioning System Module (GPSM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test Y: DTC U0100:00

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
U0100:00	Lost Communication with ECM/PCM "A": No Sub Type Information	Set by the <u>GPSM</u> when network messages are missing from the <u>PCM</u> for more than 5 seconds with the ignition in RUN.

PINPOINT TEST Y : DTC U0100:00

Y1 VERIFY THE CUSTOMER'S CONCERN

- Ignition ON.
- Verify if there is an observable symptom present.

Is an observable symptom present?

Yes	GO to Y2 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

Y2 RECHECK FOR DTC U0100:00

- Ignition ON.
- Using a diagnostic scan tool, clear all CMDTCs .
- Ignition OFF.
- Ignition ON.
- Wait at least 10 seconds.
- Using a diagnostic scan tool, retrieve all CMDTCs .

Is DTC U0100:00 set in the GPSM ?

Yes	GO to Y3 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

Y3 CHECK THE PCM FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS

- Using a diagnostic scan tool, perform the **PCM** self-test.

Are any low or high voltage DTCS recorded in the **PCM** ?

Yes	REFER to PCM DTC Chart in Section 303-14.
No	GO to Y4 .

Y4 CHECK THE GPSM FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS

- Using a diagnostic scan tool, perform the **GPSM** self-test.

Are any low or high voltage DTCS recorded in the **GPSM** ?

Yes	For DTC U3003:16, GO to Pinpoint Test AF . For DTC U3003:17, GO to Pinpoint Test AG .
No	GO to Y5 .

Y5 CHECK FOR CORRECT PCM OPERATION

- Ignition OFF.
- Disconnect and inspect all the **PCM** connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **PCM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new PCM . REFER to Section 303-14.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test Z: DTC U0140:00

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
U0140:00	Lost Communication With Body Control Module: No Sub Type Information	Set by the APIM and GPSM when network messages are missing from the BCM for more than 5 seconds with the ignition in RUN.

PINPOINT TEST Z : DTC U0140:00

Z1 VERIFY THE CUSTOMER'S CONCERN

- Ignition ON.
- Verify if there is an observable symptom present.

Is an observable symptom present?

Yes	GO to Z2 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

Z2 RECHECK FOR DTC U0140:00

- Ignition ON.
- Using a diagnostic scan tool, clear all ~~CMDTCs~~ .
- Ignition OFF.
- Ignition ON.
- Wait at least 10 seconds.
- Using a diagnostic scan tool, retrieve all ~~CMDTCs~~ .

Is DTC U0140:00 set in the ~~APIM~~ or ~~GPSM~~ ?

Yes	GO to Z3 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

Z3 CHECK THE BCM FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS

- Using a diagnostic scan tool, perform the ~~BCM~~ self-test.

Are any low or high voltage DTCs recorded in the ~~BCM~~ ?

Yes	REFER to BCM DTC Chart in Section 419-10.
No	GO to Z4 .

Z4 CHECK FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS

- Using a diagnostic scan tool, perform the self-test for the following modules:
 - APIM
 - GPSM

Are low or high voltage DTCS recorded in the APIM or GPSM ?

Yes	For DTC U3003:16, GO to Pinpoint Test AF . For DTC U3003:17, GO to Pinpoint Test AG .
No	GO to Z5 .

Z5 CHECK FOR DTC U0140:00 SET IN MULTIPLE MODULES

- Review the recorded results from the self-test.

Is DTC U0140:00 set in more than one audio system module?

Yes	GO to Z6 .
No	If there is an observable symptom, GO to Symptom Chart — General Audio System to diagnose the observed symptom. If there is no observable symptom, the system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

Z6 CHECK FOR CORRECT BCM OPERATION

- Ignition OFF.
- Disconnect and inspect all the **BCM** connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **BCM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. .
No	The system is operating correctly at this time. The concern may have been caused by module

	connections. ADDRESS the root cause of any connector or pin issues.
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Pinpoint Test AA: DTC U0155:00

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
U0155:00	Lost Communication With Instrument Panel Cluster (IPC) Control Module: No Sub Type Information	Set by the ACM , APIM , FCDIM , FCIM , and GPSM when network messages are missing from the IPC for more than 5 seconds with the ignition in RUN.

PINPOINT TEST AA : DTC U0155:00

AA1 VERIFY THE CUSTOMER'S CONCERN	
- Ignition ON. - Verify if there is an observable symptom present.	
Is an observable symptom present?	
Yes	GO to AA2 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.
AA2 RECHECK FOR DTC U0155:00	
- Ignition ON. - Using a diagnostic scan tool, clear all CMDTCs . - Ignition OFF. - Ignition ON. - Wait at least 10 seconds. - Using a diagnostic scan tool, retrieve all CMDTCs .	
Is DTC U0155:00 set in the ACM , APIM , FCDIM , FCIM , or GPSM ?	
Yes	GO to AA3 .

No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.
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AA3 CHECK THE IPC FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS

- Using a diagnostic scan tool, perform the **IPC** self-test.

Are any low or high voltage DTCS recorded in the **IPC** ?

Yes	REFER to IPC DTC Chart in Section 413-01.
No	GO to AA4 .

AA4 CHECK FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS

- Using a diagnostic scan tool, perform the self-test for the following modules:
 - ACM
 - APIM
 - ECDIM
 - ECIM
 - GPSM

Are low or high voltage DTCS recorded in the **ACM** , **APIM** , **ECDIM** , **ECIM** , or **GPSM** ?

Yes	For DTC U3003:16, GO to Pinpoint Test AF . For DTC U3003:17, GO to Pinpoint Test AG .
No	GO to AA5 .

AA5 CHECK FOR DTC U0155:00 SET IN MULTIPLE MODULES

- Review the recorded results from the self-test.

Is DTC **U0155:00** set in more than one audio system module?

Yes	GO to AA6 .
No	If there is an observable symptom, GO to Symptom Chart — General Audio System to diagnose the observed symptom. If there is no observable symptom, the system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AA6 CHECK FOR CORRECT IPC OPERATION

- Ignition OFF.
- Disconnect and inspect the **IPC** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)

- Reconnect the **IPC** connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new IPC . REFER to Section 413-01.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test AB: DTC U0184:00

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
U0184:00	Lost Communication With Radio: No Sub Type Information	Set by the FCDIM when network messages are missing from the ACM for more than 5 seconds with the ignition in RUN.

PINPOINT TEST AB : DTC U0184:00

AB1 VERIFY THE CUSTOMER'S CONCERN

- Ignition ON.
- Verify if there is an observable symptom present.

Is an observable symptom present?

Yes	GO to AB2 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AB2 RECHECK FOR DTC U0184:00

- Ignition ON.
- Using a diagnostic scan tool, clear all **CMDTCs** .
- Ignition OFF.

- Ignition ON.
- Wait at least 10 seconds.
- Using a diagnostic scan tool, retrieve all **CMDTCs** .

Is DTC U0184:00 set in the **FCDIM ?**

Yes	GO to AB3 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AB3 CHECK FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS

- Using a diagnostic scan tool, perform the self-test for the following modules:
 - **ACM**
 - **FCDIM**

Are low or high voltage DTCs recorded in the **ACM or **FCDIM** ?**

Yes	For DTC U3003:16, GO to Pinpoint Test AF . For DTC U3003:17, GO to Pinpoint Test AG .
No	GO to AB4 .

AB4 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the **ACM** connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **ACM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test AC: DTC U0253:00

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
U0253:00	Lost Communication With Accessory Protocol Interface Module: No Sub Type Information	Set by the ACM and ECDIM when network messages are missing from the APIM for more than 5 seconds with the ignition in RUN.

PINPOINT TEST AC : DTC U0253:00

AC1 VERIFY THE CUSTOMER'S CONCERN

- Ignition ON.
 - Verify if there is an observable symptom present.
- Is an observable symptom present?**

Yes	GO to AC2 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AC2 RECHECK FOR DTC U0253:00

- Ignition ON.
 - Using a diagnostic scan tool, clear all CMDTCs .
 - Ignition OFF.
 - Ignition ON.
 - Wait at least 10 seconds.
 - Using a diagnostic scan tool, retrieve all CMDTCs .
- Is DTC U0253:00 set in the ACM or ECDIM ?**

Yes	GO to AC3 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AC3 CHECK FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS

- Using a diagnostic scan tool, perform the self-test for the following modules:

- ACM
- APIM
- ECDIM

Are low or high voltage DTCs recorded in the **ACM** , **APIM** , or **ECDIM** ?

Yes	For DTC U3003:16, GO to Pinpoint Test AF . For DTC U3003:17, GO to Pinpoint Test AG .
No	GO to AC4 .

AC4 CHECK FOR DTC U0253:00 SET IN MULTIPLE MODULES

- Review the recorded results from the self-test.

Is DTC **U0253:00** set in more than one audio system module?

Yes	GO to AC5 .
No	If there is an observable symptom, GO to Symptom Chart — General Audio System to diagnose the observed symptom. If there is no observable symptom, the system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AC5 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.
- Disconnect and inspect the **APIM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **APIM** connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the **APIM** to re-initialize.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test AD: DTC U0256:00

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
U0256:00	Lost Communication With Front Controls Interface Module "A": No Sub Type Information	Set by the ACM and ECDIM when network messages are missing from the ECIM for more than 5 seconds with the ignition in the RUN position.

PINPOINT TEST AD : DTC U0256:00

AD1 VERIFY THE CUSTOMER'S CONCERN

- Ignition ON.
- Verify if there is an observable symptom present.

Is an observable symptom present?

Yes	GO to AD2 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AD2 RECHECK FOR DTC U0256:00

- Ignition ON.
- Using a diagnostic scan tool, clear all ~~CMDTCs~~ .
- Ignition OFF.
- Ignition ON.
- Wait at least 10 seconds.
- Using a diagnostic scan tool, retrieve all ~~CMDTCs~~ .

Is DTC U0256:00 set in the ~~ACM~~ or ~~ECDIM~~ ?

Yes	GO to AD3 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AD3 CHECK FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS

- Using a diagnostic scan tool, perform the self-test for the following modules:

- ACM
- ECDIM
- FCIM

Are low or high voltage DTCs recorded in the **ACM** , **ECDIM** , or **FCIM** ?

Yes	For DTC U3003:16, GO to Pinpoint Test AF . For DTC U3003:17, GO to Pinpoint Test AG .
No	GO to AD4 .

AD4 CHECK FOR DTC U0256:00 SET IN MULTIPLE MODULES

- Review the recorded results from the self-test.

Is DTC **U0256:00** set in more than one audio system module?

Yes	GO to AD5 .
No	If there is an observable symptom, GO to Symptom Chart — General Audio System to diagnose the observed symptom. If there is no observable symptom, the system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AD5 CHECK FOR CORRECT FCIM OPERATION

- Ignition OFF.
- Disconnect and inspect the **FCIM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **FCIM** connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new FCIM . REFER to Front Controls Interface Module (FCIM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test AE: DTC U0257:00

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
U0257:00	Lost Communication With Front Controls / Display Interface Module: No Sub Type Information	Set by the ACM , APIM , and ECIM when Infotainment Controller Area Network (I-CAN) network messages are missing from the FCDIM for more than 5 seconds with the ignition in RUN.

PINPOINT TEST AE : DTC U0257:00

AE1 VERIFY THE CUSTOMER'S CONCERN	
- Ignition ON. - Verify if there is an observable symptom present.	
Is an observable symptom present?	
Yes	GO to AE2 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.
AE2 RECHECK FOR DTC U0257:00	
- Ignition ON. - Using a diagnostic scan tool, clear all CMDTCs . - Ignition OFF. - Ignition ON. - Wait at least 10 seconds. - Using a diagnostic scan tool, retrieve all CMDTCs .	
Is DTC U0257:00 set in the ACM , APIM , or ECIM ?	
Yes	GO to AE3 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.
AE3 CHECK FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS	
- Using a diagnostic scan tool, perform the self-test for the following modules: ACM APIM	

- FCDIM
- FCIM

Are low or high voltage DTCs recorded in the ACM , APIM , FCDIM , or FCIM ?

Yes	For DTC U3003:16, GO to Pinpoint Test AF . For DTC U3003:17, GO to Pinpoint Test AG .
No	GO to AE4 .

AE4 CHECK FOR DTC U0257:00 SET IN MULTIPLE MODULES

- Review the recorded results from the self-test.

Is DTC U0257:00 set in more than one audio system module?

Yes	GO to AE5 .
No	If there is an observable symptom, GO to Symptom Chart — General Audio System to diagnose the observed symptom. If there is no observable symptom, the system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AE5 CHECK FOR CORRECT FCDIM OPERATION

- Ignition OFF.
- Disconnect and inspect the ~~FCDIM~~ connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the ~~FCDIM~~ connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new FCDIM . REFER to Front Controls/Display Interface Module (FCDIM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test AF: DTC U3003:16

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
U3003:16	Battery Voltage: Circuit Voltage Below Threshold	Set by the ACM , APIM , FCDIM , FCIM , and GPSM when the supply voltage falls below 10 volts for at least 10 seconds during normal operation or for more than 250 milliseconds during the self-test.

PINPOINT TEST AF : DTC U3003:16

AF1 RECHECK FOR DTC U3003:16	
- Ignition ON. - Using a diagnostic scan tool, clear the DTCs for the module in question. - Wait at least 15 seconds. - Using a diagnostic scan tool, repeat the self-test for the module in question.	
Is DTC U3003:16 still present?	
Yes	GO to AF2 .
No	The system is operating correctly at this time. The DTC may have been set due to a previous low battery voltage condition.

AF2 CHECK FOR CHARGING SYSTEM DTCS IN THE PCM	
Using a diagnostic scan tool, perform the PCM self-test.	
Are any voltage-related DTCs set in the PCM ?	
Yes	REFER to PCM DTC Chart in Section 303-14.
No	GO to AF3 .

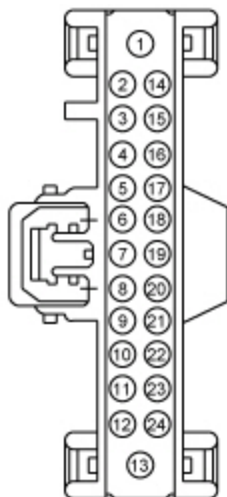
AF3 CHECK THE BATTERY CONDITION AND STATE OF CHARGE	
Check the battery condition and verify that the battery is fully charged. REFER to Battery Condition Test in Section 414-01.	
Is the battery OK and fully charged?	

Yes	GO to AF4 .
No	REFER to the Symptom Chart in Section 414-00.

AF4 CHECK THE MODULE VOLTAGE SUPPLY

- Ignition OFF.
- Measure and record the voltage at the battery.
- Disconnect the audio system module setting DTC U3003:16.
- Ignition ON.
- Measure the voltage between the module setting DTC U3003:16 and ground.
- For the ~~ACM~~ , measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C240A Pin 1	SBP29 (GY/RD)	—	Ground



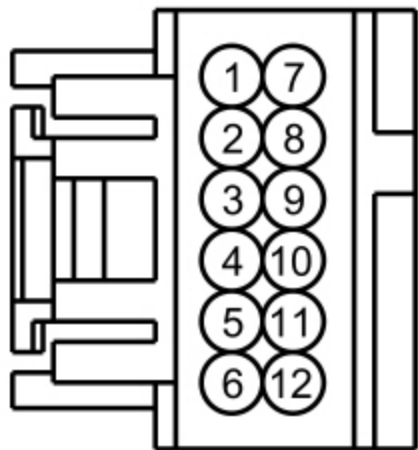
-
- For the ~~APIM~~ , measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2383 Pin 1	SBP02 (YE/RD)	—	Ground



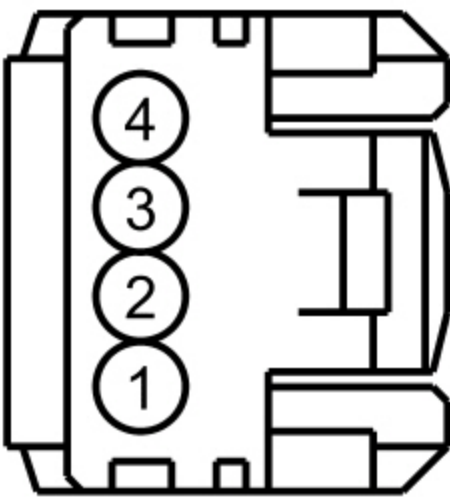
-
- For the **ECDIM** , measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2123 Pin 1	SBP09 (RD)	—	Ground



-
- For the **ECIM** , measure the **voltage** between:

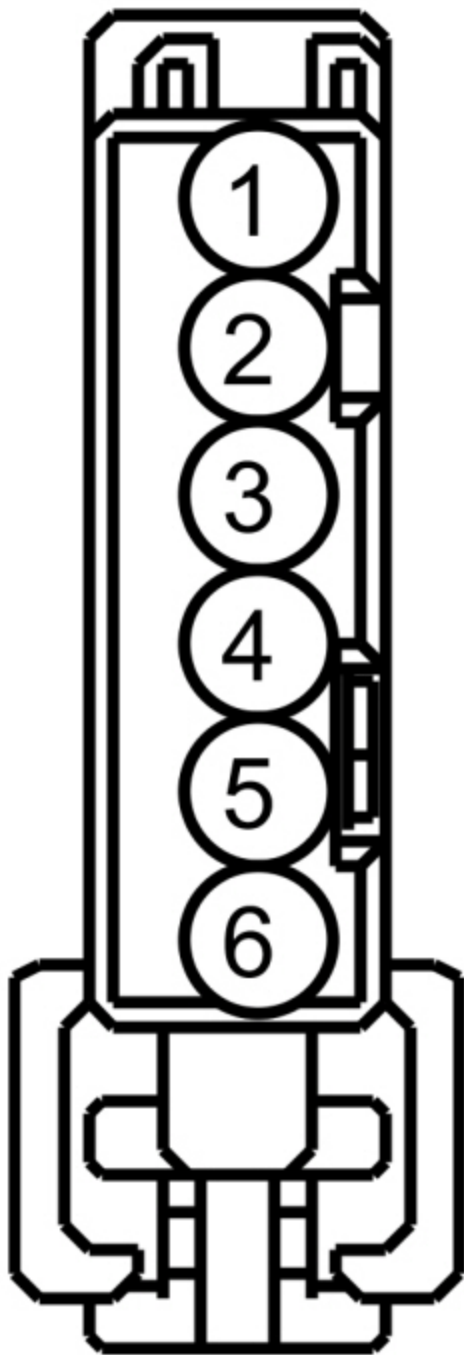
Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2402 Pin 1	SBP09 (RD)	—	Ground



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- For the GPSM , measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2398 Pin 1	SBP09 (RD)	—	Ground



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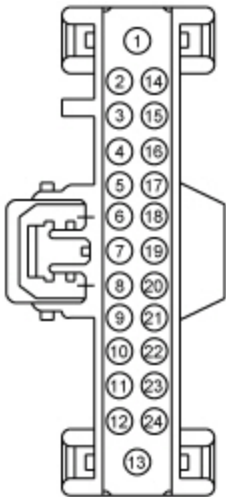
Is the voltage within 0.2 volts of the recorded battery voltage?

Yes	GO to AF5 .
No	REPAIR the circuit for high resistance.

AF5 CHECK THE SUSPECT MODULE GROUND CIRCUIT FOR CONTINUITY

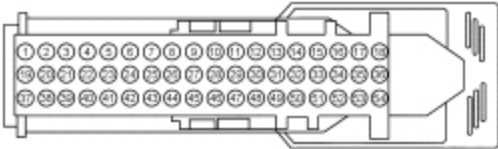
- Measure the voltage at the suspect module.
- For the ACM , measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C240A Pin 1	SBP29 (GY/RD)	C240A Pin 13	GD114 (BK/BU)



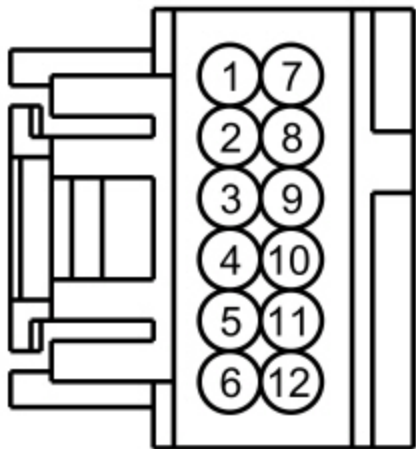
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- For the ~~APIM~~ , measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2383 Pin 1	SBP02 (YE/RD)	C2383 Pin 37	GD114 (BK/BU)
		C2383 Pin 38	GD114 (BK/BU)



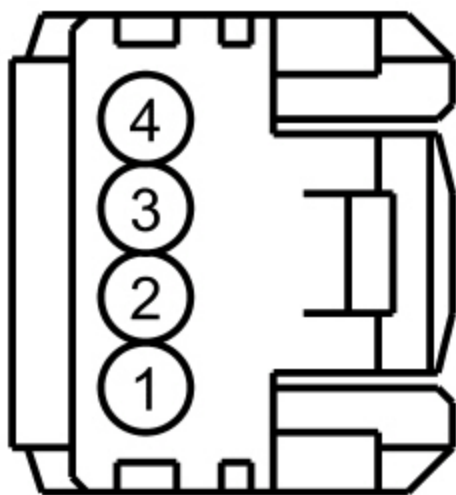
-
- For the ~~FCDIM~~ , measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2123 Pin 1	SBP09 (RD)	C2123 Pin 9	GD114 (BK/BU)



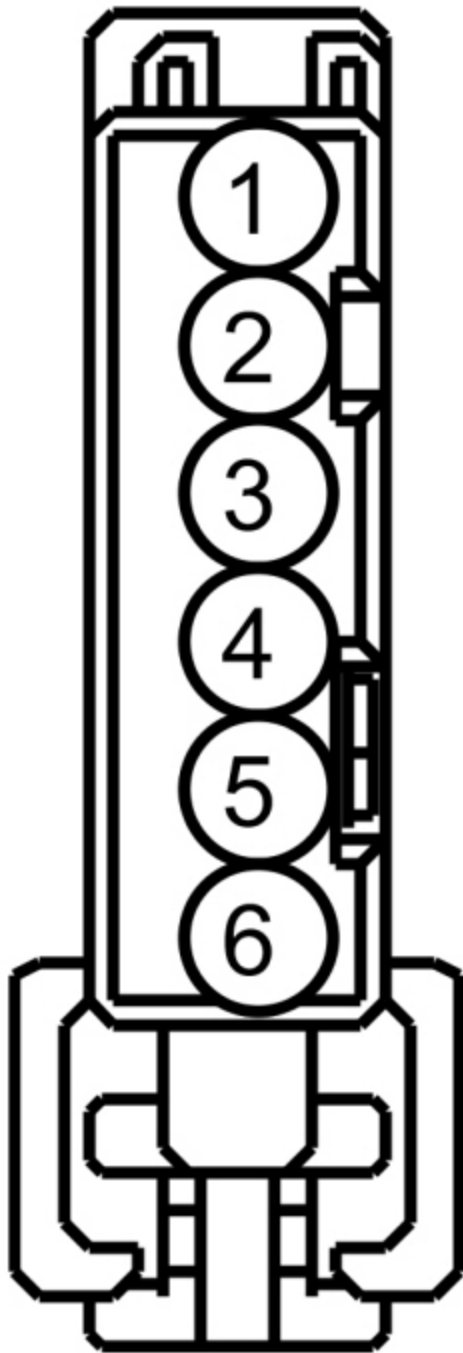
-
- For the ECIM , measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2402 Pin 1	SBP09 (RD)	C2402 Pin 4	GD133 (BK)



-
- For the GPSM , measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2398 Pin 1	SBP09 (RD)	C2398 Pin 6	GD114 (BK/BU)



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Is the suspect module voltage greater than 11 volts?

Yes	For the <u>ACM</u> , GO to AF6 . For the <u>APIM</u> , GO to AF7 . For the <u>FCDIM</u> , GO to AF8 . For the <u>FCIM</u> , GO to AF9 . For the <u>GPSM</u> , GO to AF10 .
No	REPAIR the circuit for high resistance.

- Ignition OFF.
- Disconnect and inspect all the **ACM** connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **ACM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

AF7 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.
- Disconnect and inspect the **APIM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **APIM** connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the **APIM** to re-initialize.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM . Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

AF8 CHECK FOR CORRECT FCDIM OPERATION

- Ignition OFF.
- Disconnect and inspect the **FCDIM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)

- Damaged or bent pins (install new terminals or pins)
- Pushed-out pins (install new pins as necessary)

- Reconnect the **FCDIM** connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new FCDIM . REFER to Front Controls/Display Interface Module (FCDIM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

AF9 CHECK FOR CORRECT FCIM OPERATION

- Ignition OFF.
- Disconnect and inspect the **FCIM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **FCIM** connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new FCIM . REFER to Front Controls Interface Module (FCIM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

AF10 CHECK FOR CORRECT GPSM OPERATION

- Ignition OFF.
- Disconnect and inspect the **GPSM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **GPSM** connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new GPSM . REFER to Global Positioning System Module (GPSM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test AG: DTC U3003:17

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
U3003:17	Battery Voltage: Circuit Voltage Above Threshold	Set by the ACM , APIM , FCDIM , FCIM , and GPSM when the supply voltage is more than 15.8 volts for more than 250 milliseconds during normal operation or the self-test.

PINPOINT TEST AG : DTC U3003:17

AG1 RECHECK FOR DTC U3003:17	
• Turn the engine off. • Ignition ON. • Using a diagnostic scan tool, clear the DTC for the module in question. • Wait at least 10 seconds. • Using a diagnostic scan tool, perform the self-test for the module in question.	
Is DTC U3003:17 present?	
Yes	GO to AG2 .
No	The system is operating correctly at this time. The DTC may have been set previously during battery charging or while jump starting the vehicle.
AG2 CHECK FOR HIGH VOLTAGE DTCS SET IN OTHER MODULES	
• Ignition ON. • Using a diagnostic scan tool, retrieve all CMDTCs .	

- Wait at least 10 seconds.
- Using a diagnostic scan tool, perform the self-test for the module in question.

Are high voltage DTCs such as B1676, P0563, U3003:17 set in more than one module?

Yes	REFER to Section 414-01 to diagnose an overcharging condition.
No	GO to AG3 .

AG3 CHECK THE BATTERY VOLTAGE

- Turn off all interior lights, exterior lights, and accessories.
- Start and run the engine at approximately 2,000 RPM for 3 minutes while monitoring the battery voltage.

Does the battery voltage rise to 15.5 volts or higher?

Yes	REFER to Section 414-01 to diagnose an overcharging condition.
No	For the ACM , GO to AG4 . For the APIM , GO to AG5 . For the FCDIM , GO to AG6 For the FCIM , GO to AG7 For the GPSM , GO to AG8 .

AG4 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the ~~ACM~~ connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the ~~ACM~~ connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

AG5 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.

- Disconnect and inspect the **APIM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **APIM** connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the **APIM** to re-initialize.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM . Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

AG6 CHECK FOR CORRECT FCDIM OPERATION

- Ignition OFF.
- Disconnect and inspect the **FCDIM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **FCDIM** connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new FCDIM . REFER to Front Controls/Display Interface Module (FCDIM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

AG7 CHECK FOR CORRECT FCIM OPERATION

- Ignition OFF.
- Disconnect and inspect the **FCIM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)

- Pushed-out pins (install new pins as necessary)
- Reconnect the **FCIM** connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new FCIM . REFER to Front Controls Interface Module (FCIM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

AG8 CHECK FOR CORRECT GPSM OPERATION

- Ignition OFF.
- Disconnect and inspect the **GPSM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **GPSM** connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new GPSM . REFER to Global Positioning System Module (GPSM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.